Manuscript: Post-transcriptional spliceosome accumulation defines a therapeutically targetable malignant cell state in glioblastoma

This document describes all supplementary figures and tables.

Figure S1 | CopyKAT purity control. (A) Aneuploid (tumor) fraction per sample estimated by copy-number inference (18 samples, 16,255 cells called). (B) WEBS distribution across LLS/LDTS/LHS within aneuploid-only cells (n = 4,421; Kruskal?Wallis p < 10?¹?), demonstrating that WEBS stratification is not an artifact of non-malignant admixture.

Figure S2 | Parameter sensitivity. (a) WEBS computed at UCell maxRank from 5% to 90% of genes vs the default (1,500 genes); WEBS is stable up to maxRank ? 0.10 and drifts at larger values, motivating the fixed maxRank = 1,500 setting. (b) Leave-one-out and leave-two-out removal of writer/eraser genes (57 perturbations; median Spearman ? = 0.95 vs default). (c) Cross-platform consistency of patient-level WEBS between Smart-seq2 and 10x in GSE131928 (? = 0.67, n = 10 shared patients).

Figure S3 | Method benchmarking and computational performance. Patient-level WEBS vs AUCell (? = 0.95), ssGSEA (? = 0.87), all-gene mean (? = 0.92); PLAGE is sign-inverted (? = ?0.97, a known PCA orientation artifact). Runtime and memory scaling at 1k?35.8k cells: WEBS completes 35,823 cells in 213 s with <2 MB incremental memory; ssGSEA exceeds available memory beyond ~25k cells.

Figure S4 | Pan-cancer scope. WEBS?survival associations across TCGA cohorts (GBM, LUAD, BRCA, SKCM): significant in LUAD (HR = 0.64, p = 0.039) but not in BRCA/SKCM/GBM-bulk, delineating the cancer-type-specific scope of the axis.

Figure S5 | Bootstrap stability of the LASSO-Cox feature selection. Selection frequency of candidate genes across 1,000 bootstrap resamples of the training cohort; HDAC1 (99.9%), LDHB (98%), VEGFA (94%), KAT8 (89%) are the most stable features.

Figure S6 | SHAP interpretability of the six-gene model. Mean |SHAP| per gene overall and by risk group; HDAC1 is the dominant contributor (protective direction, consistent with its eraser role).

Figure S7 | Spliceosome specificity control. Layer discordance (protein ? ? mRNA ? vs WEBS) for the spliceosome (0.81) against 10,000 random gene sets matched for proteome abundance (empirical p = 0.22), demonstrating that partial discordance is generic to abundant complexes; spliceosome specificity therefore rests on direction-uniform up-regulation and functional evidence rather than discordance magnitude. The proteasome is shown as a secondary housekeeping-complex case (0.77).

Figure S8 | Spliceosome acetylation-site heatmap. Top 40 differentially acetylated spliceosome sites (limma, FDR < 0.05), samples ordered by WEBS group; site-level acetylation is enriched in the writer-dominant state (OR = 7.36, p = 1.3 × 10?¹?).

Figure S9 | MOFA2 four-view integration (CPTAC). (a) Factor?WEBS Spearman correlations across 10 latent factors; Factor 2 covaries with WEBS (? = ?0.637, p < 10?¹²). (b) Factor 2 vs protein-level WEBS across 99 CPTAC-GBM samples.

Figure S10 | Spatial autocorrelation of WEBS. Moran's I per Visium slice with 1,000-permutation p-values; significant clustering in GBM3, GBM5_2, and GBM4.

Figure S11 | GSE83300 external validation. Kaplan?Meier for the six-gene signature in GSE83300 (n = 50, 39 events): HR = 2.76 with concordant direction but p = 0.74, consistent with limited power at this sample size; reported transparently.

Figure S12 | Risk × treatment interaction. ?log10(p) of the signature-by-treatment interaction term in Cox models (CGGA-693 and CGGA-325), for temozolomide and radiotherapy. Significant interactions in both cohorts indicate treatment-context-dependent prognostic value.

Figure S13 | Joint MGMT × signature stratification. Kaplan?Meier curves of four strata (MGMT methylated/unmethylated × high/low risk) in CGGA-325 and CGGA-693 (both $p < 10?^{13}$).

Figure S14 | GLASS longitudinal WEBS. WEBS in paired primary and first-recurrence glioma samples (n = 179 pairs; paired Wilcoxon p = 0.09): no systematic change at recurrence.

Figure S15 | Meta-analysis of the six-gene signature. Fixed-effect meta-analysis across TCGA-GBM, CGGA-693, and CGGA-325 (pooled HR = 1.35, p = 0.0028).

Figure S16 | Similarity network fusion (SNF). Consensus clustering across CPTAC proteome, acetylome, and transcriptome; clusters recapitulate WEBS stratification (Kruskal?Wallis p = 7.1 × 10??).

Figure S17 | WGCNA module?WEBS association (CPTAC proteome). The most WEBS-correlated module (ME3, r = 0.72, $q = 3.7 \times 10?^{1}?$) is neural-lineage- rather than spliceosome-enriched; the spliceosome signal is distributed, not module-confined (negative control).

Figure S18 | STRING protein?protein interaction network. The spliceosome protein set forms a denser network than chance expectation (121 nodes, 3,349 edges, enrichment p < 10???; hubs SNW1, SRSF1, SF3B1).

Figure S19 | Pathway-level SHAP aggregation. SHAP values of the six-gene Cox model aggregated to Hallmark pathways; hypoxia and glycolysis programs dominate risk contribution.

Figure S20 | Genome-wide discordance tail. Distribution of per-gene protein?mRNA discordance; the spliceosome lies in the extreme tail (p = 4.7 × 10???).

Figure S21 | Pseudoreplication control. Per-patient Spearman correlations for the two key single-cell associations (WEBS vs Hallmark Hypoxia; WEBS vs Slingshot pseudotime), each point = one patient. Hypoxia decoupling is confirmed at patient level (median ? ? 0, one-sample Wilcoxon p = 1); the pseudotime trend attenuates at patient level (p = 0.22) and is presented as cell-level supportive evidence. Pseudobulk mixed-effects models (patient as random intercept) confirm the Neftel-state gradient (p = 7.4 × 10??).

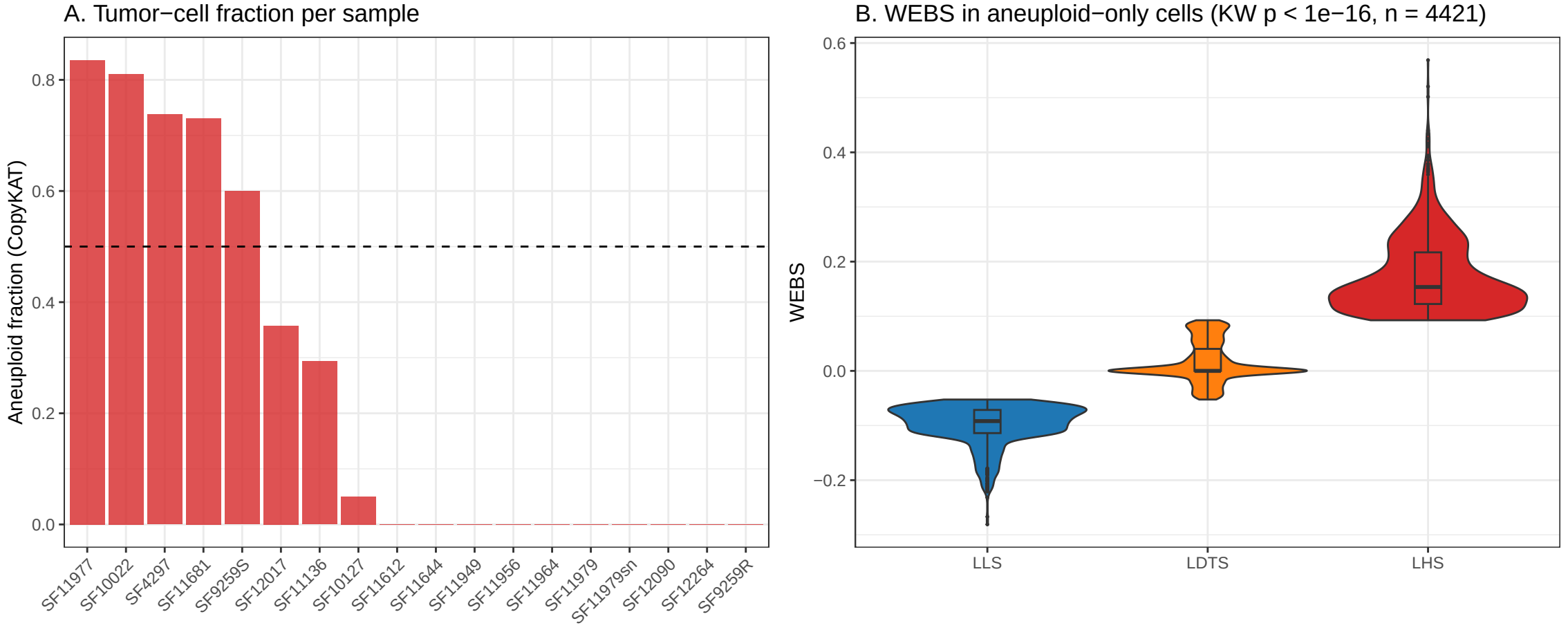
Figure S22 | Per-gene RNA vs protein correlation with WEBS. Each point is one spliceosome gene (n = 109 with both layers); 107/109 show stronger protein- than mRNA-layer correlation. Bootstrap of the pathway-level ?? (2,000 resamples): 95% CI 0.60?1.03.

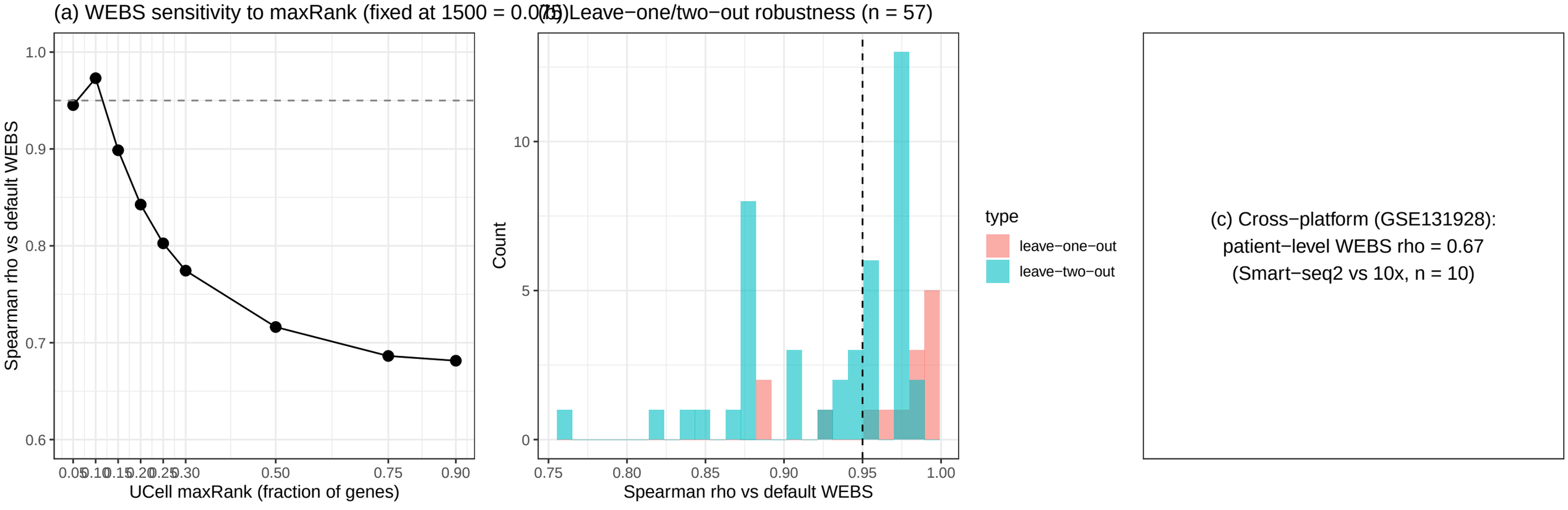
Figure S23 | Cross-cohort replication of WEBS-associated splicing events. Effect sizes (PSI vs WEBS Spearman ?) in TCGA-GBM vs TCGA-LGG for the 4,285 testable GBM-identified events; 83% sign-concordant (binomial $p < 10?^{1}?$), effect-size correlation ? = 0.70.

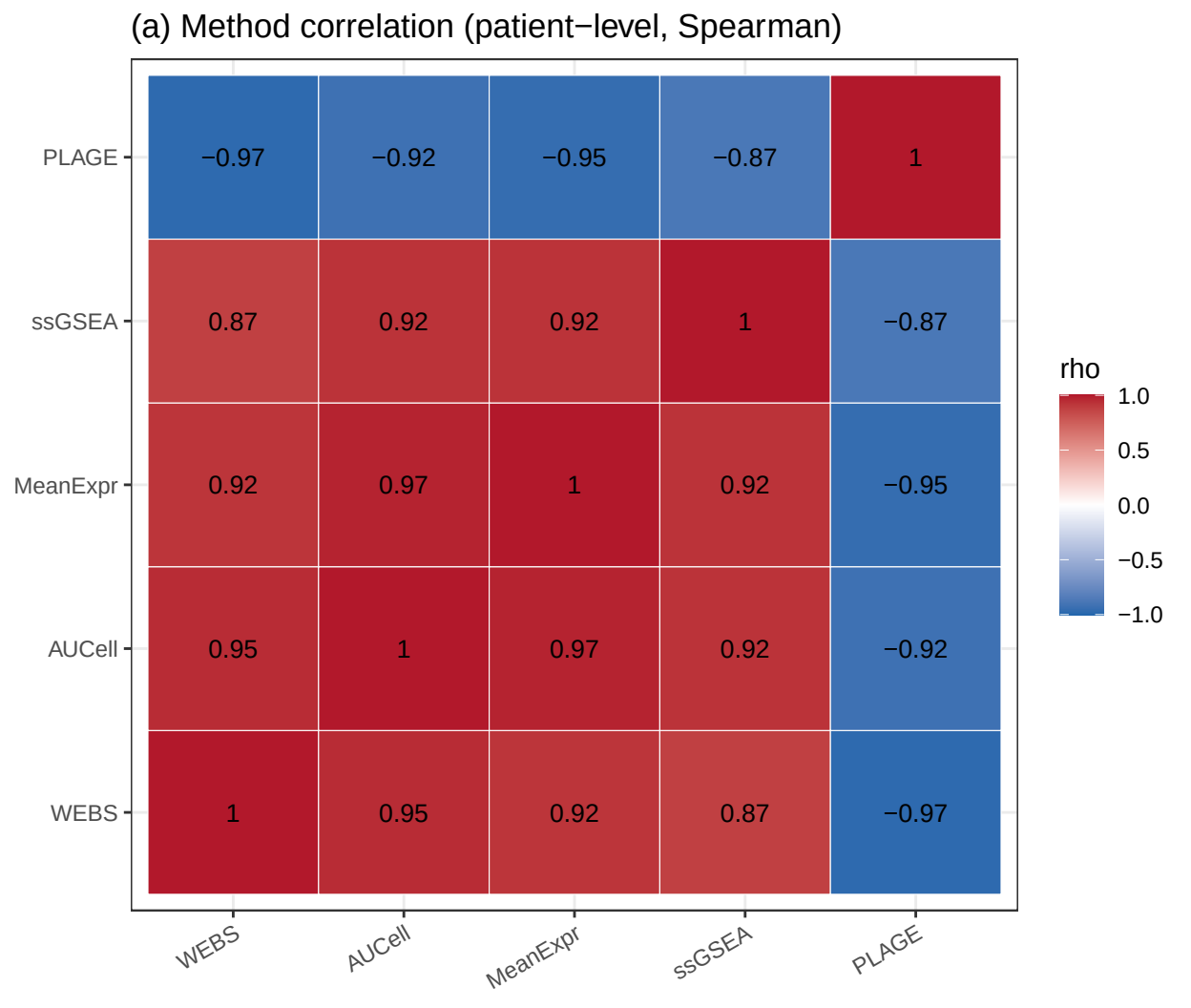
Sheet	Content
S_CPTAC_WEBS	Protein-level WEBS and group assignment for 99 CPTAC-GBM samples
S_CPTAC_limma	Full limma differential abundance results (11,141 proteins, LHS vs LLS)
S_spliceosome	Spliceosome protein subset results (111 quantified, 107 up-regulated)
S_acetyl_sites	Site-level acetylome differential analysis (8,350 sites)
S_acetyl_corr	Per-site acetylation?WEBS correlations
S_bulk_survival	WEBS survival analysis across bulk cohorts
S_signature_valid	Six-gene signature validation across cohorts (fixed coefficients)
S_bootstrap	Bootstrap selection frequencies (100 resamples)
S_SHAP	SHAP values summary per gene and risk group
S_pancancer	Pan-cancer WEBS?survival summary (4 TCGA cohorts)
S_benchmark	Method correlation matrix (WEBS/AUCell/ssGSEA/PLAGE/mean)
S_performance	Computational runtime and memory at 1k?35.8k cells
S_sensitivity_maxrank	UCell maxRank sweep
S_sensitivity_loo	Leave-one/two-out robustness (57 perturbations)
S_specificity	Spliceosome vs random-pathway discordance control
S_spatial	Moran's I / Geary's c spatial autocorrelation per slice
S_MOFA2	MOFA2 factor?WEBS correlations (4-view CPTAC integration)
S_DepMap	WEBS?dependency correlations for candidate targets
S_GDSC2_drugs	WEBS?sensitivity correlations across 286 drugs
S_GDSC2_HDAC	HDAC-inhibitor subset results
S_combinations	Bliss-independence combination predictions
S_CMap	Drugs targeting LHS-up proteins with WEBS-linked sensitivity
S_CellChat	Pathway-level information flow LLS vs LHS
S_MIF_perm	MIF?CD74/CXCR4 permutation test
S_docking	Molecular docking binding energies (4 target?ligand pairs)
S_ablation	Ablation study: leave-one-out and single-gene C-indices of the six-gene signature (TCGA-GBM)
S_signature_performance	Six-gene signature performance across all cohorts: per-SD HR (95% CI), C-index, time-dependent AUC (12/36/60 months)
S_threshold_sensitivity	Spliceosome up-regulation robustness across WEBS stratification thresholds (109?111/111 up-regulated)

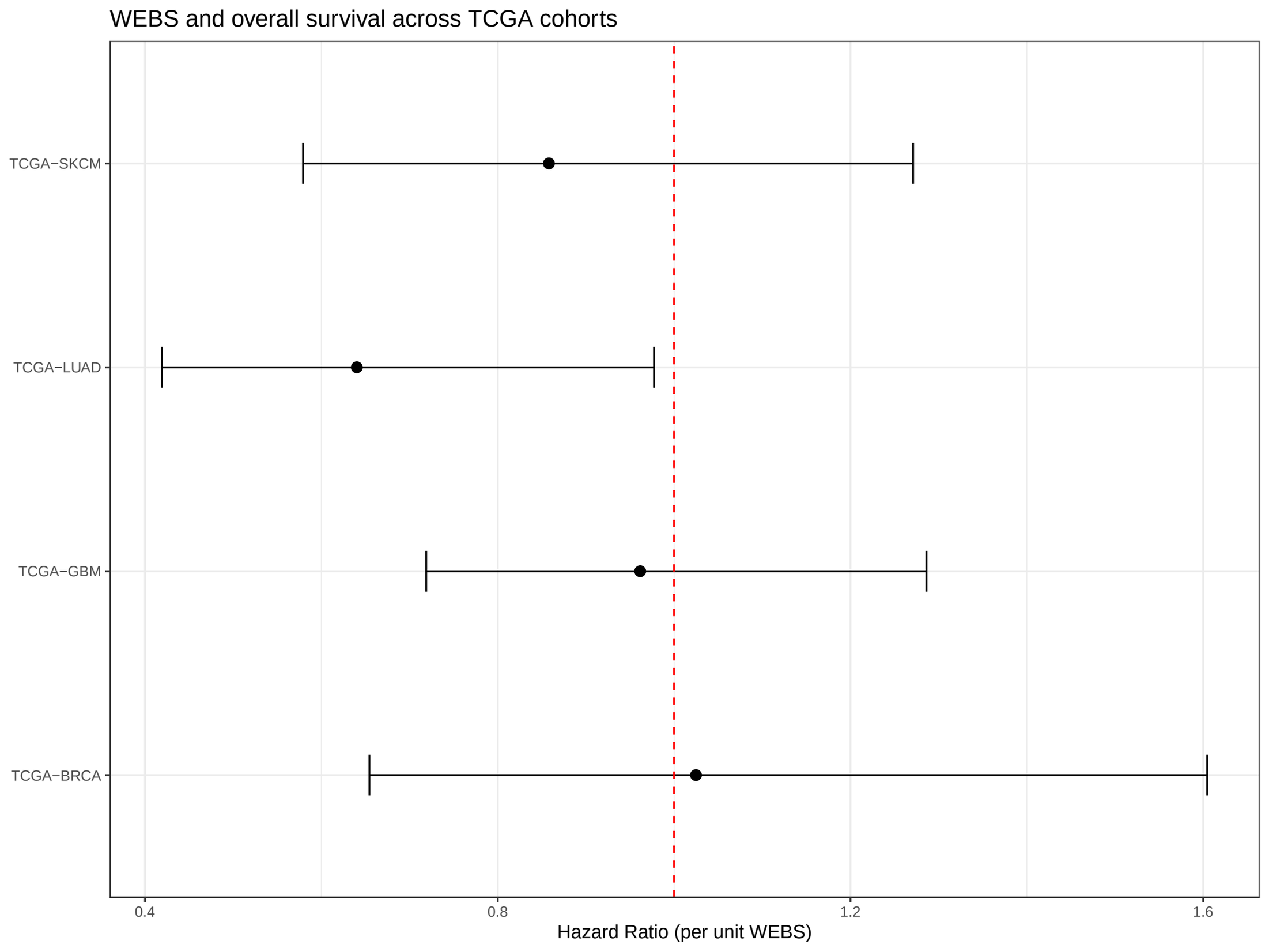
- snATAC-seq: fragment files were not deposited for GSE138794; accessibility analysis is therefore limited to peak-matrix promoter aggregation (TSS ± 2 kb), and nucleosome/TSS quality metrics could not be computed.
- RNA velocity: spliced/unspliced matrices are not deposited for GSE138794; trajectory evidence relies on Slingshot pseudotime.

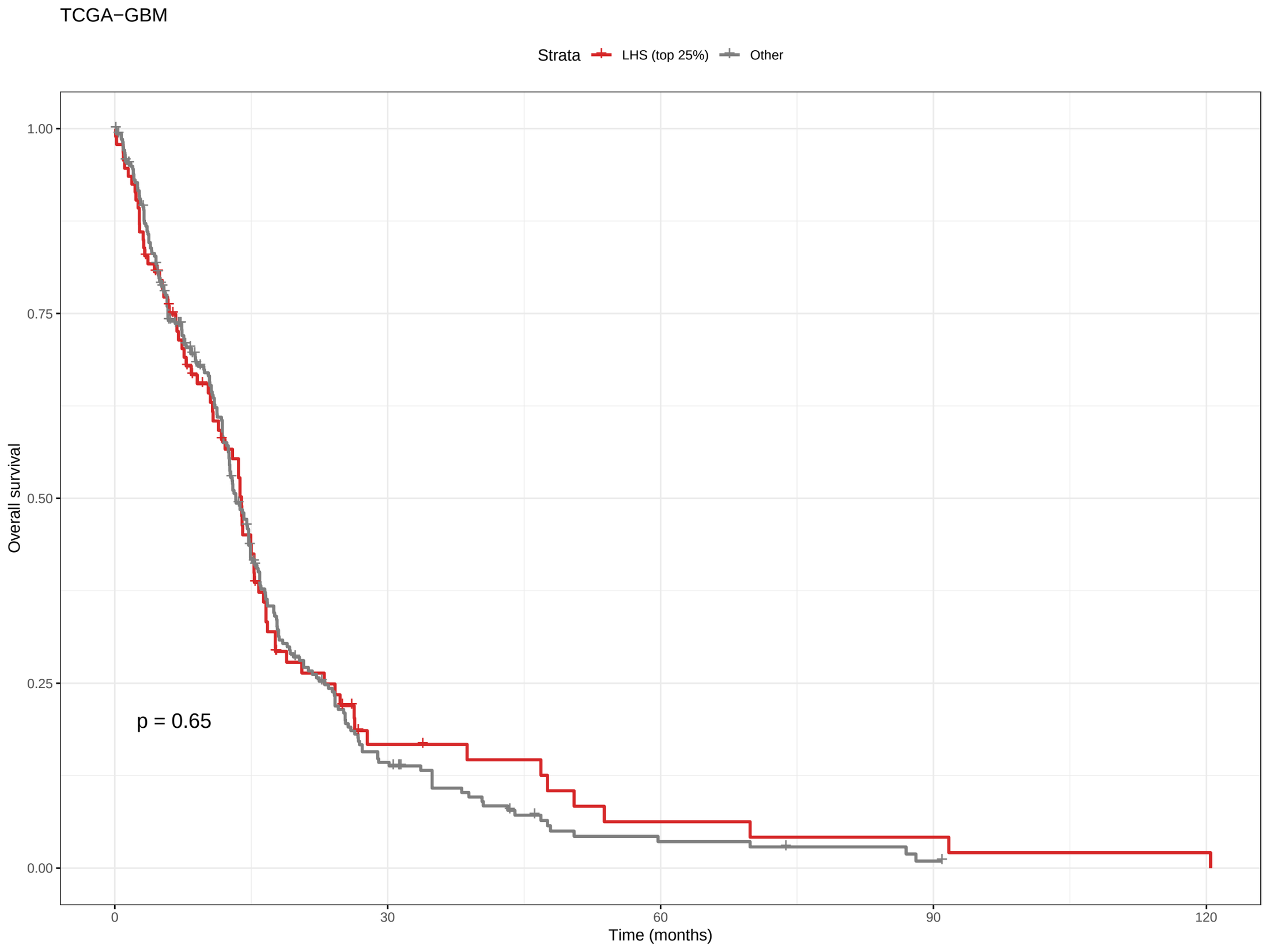
- REMBRANDT / GSE13041: these cohorts lack survival fields in their GEO series matrices and were omitted without affecting central claims. GLASS data were accessed via Synapse (syn17038081) under the consortium's data-use terms and are included for validation and longitudinal analyses (main text Figs. 13?14).
- RCTD deconvolution: spatial deconvolution was performed with Seurat label transfer; RCTD was attempted but the package failed at its internal caching step on Windows (choose_sigma_c); results were concordant with label-transfer deconvolution.

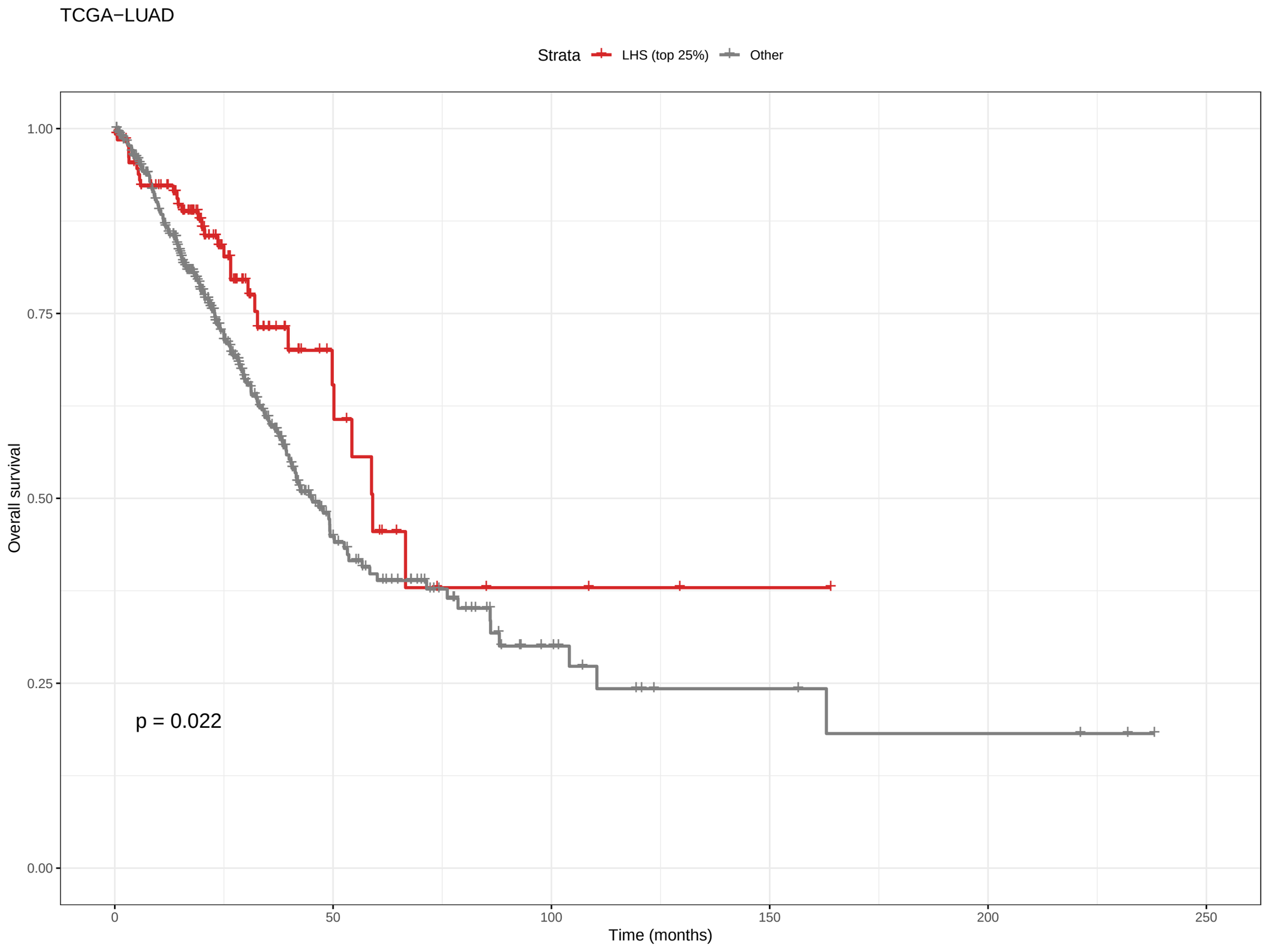

A. Tumor−cell fraction per sample
Aneuploid fraction (CopyKAT)
0.0
0.2
0.4
0.6
0.8
SF11977
SF10022
SF4297
SF11681
SF9259S
SF12017
SF11136
SF10127
SF11612
SF11644
SF11949
SF11956
SF11964
SF11979
SF11979sn
SF12090
SF12264
SF9259R
B. WEBS in aneuploid−only cells (KW p < 1e−16, n = 4421)
WEBS
−0.2
0.0
0.2
0.4
0.6
LLS
LDTS
LHS

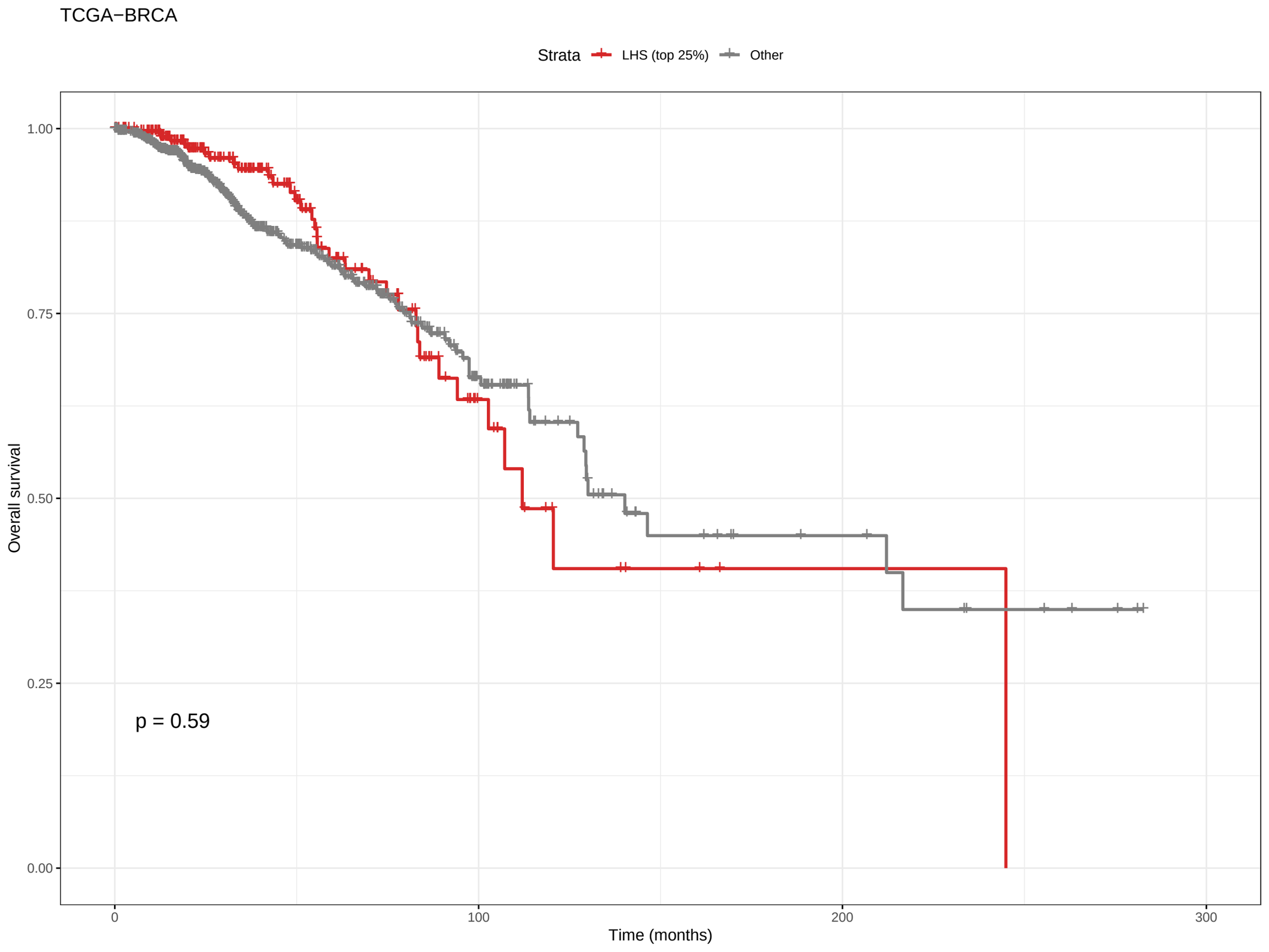

(a) WEBS sensitivity to maxRank (fixed at 1500 = 0.075)
Spearman rho vs default WEBS
1.0
0.9
0.8
0.7
0.6
0.05
0.10
0.15
0.20
0.25
0.30
0.50
0.75
0.90
UCell maxRank (fraction of genes)
(b) Leave−one/two−out robustness (n = 57)
Count
10
5
0
0.75
0.80
0.85
0.90
0.95
1.00
Spearman rho vs default WEBS
type
leave−one−out
leave−two−out
(c) Cross−platform (GSE131928):
patient−level WEBS rho = 0.67
(Smart−seq2 vs 10x, n = 10)

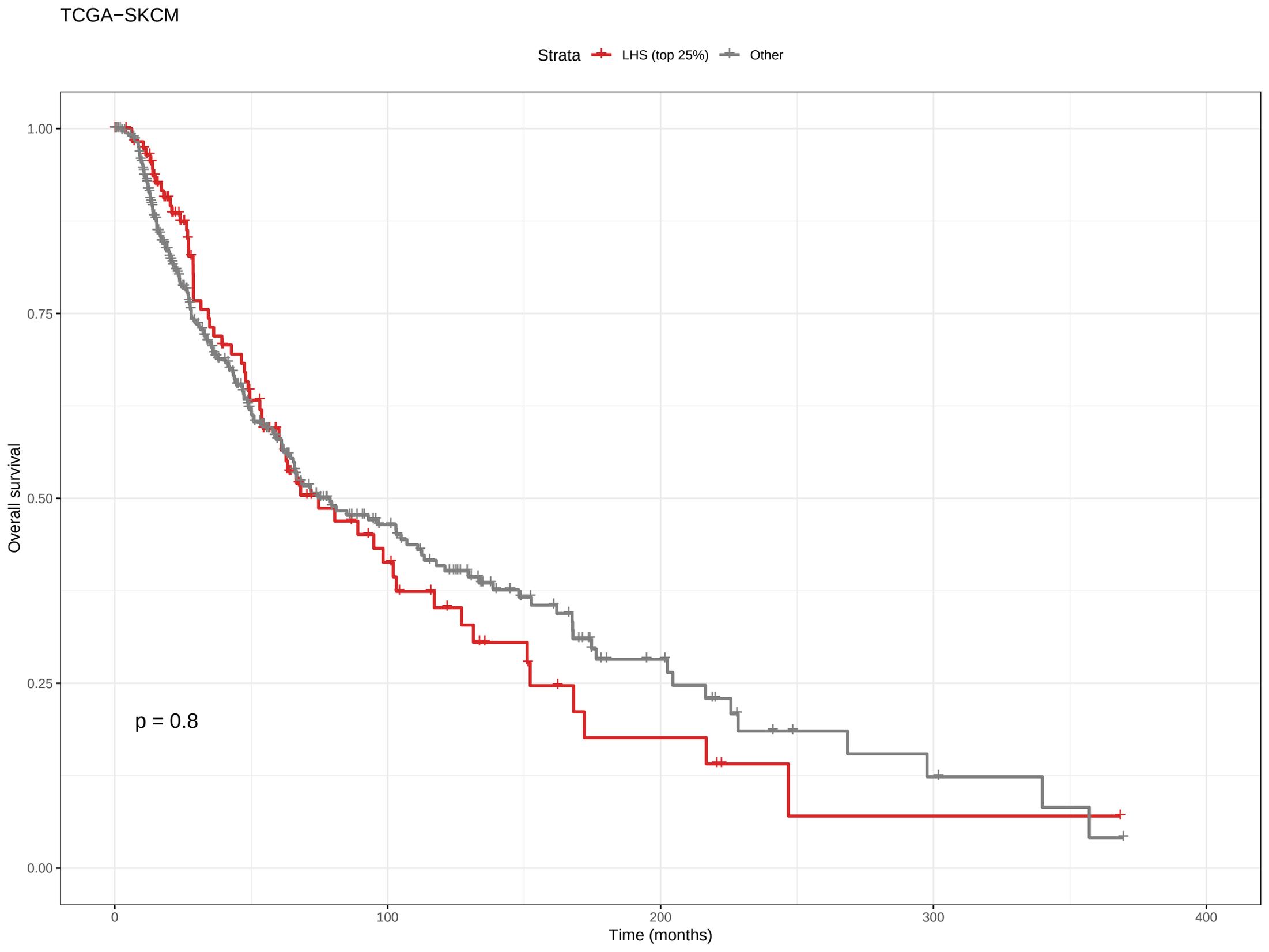

(a) Method correlation (patient−level, Spearman)
PLAGE
ssGSEA
MeanExpr
AUCell
WEBS
−0.97
−0.92
−0.95
−0.87
1
0.87
0.92
0.92
1
−0.87
0.92
0.97
1
0.92
−0.95
0.95
1
0.97
0.92
−0.92
1
0.95
0.92
0.87
−0.97
WEBS
AUCell
MeanExpr
ssGSEA
PLAGE
rho
1.0
0.5
0.0
−0.5
−1.0

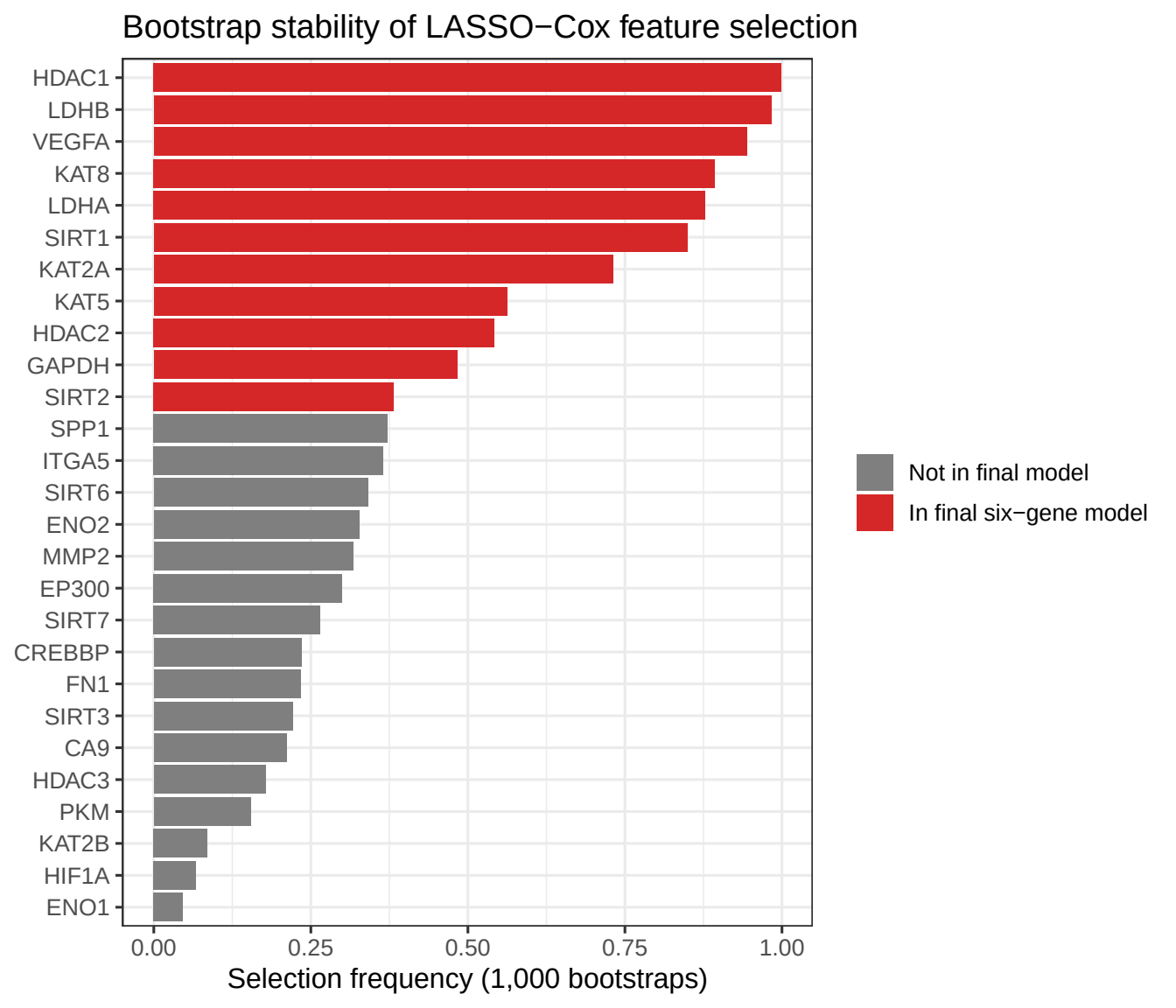

WEBS and overall survival across TCGA cohorts
TCGA−SKCM
TCGA−LUAD
TCGA−GBM
TCGA−BRCA
0.4
0.8
1.2
1.6
Hazard Ratio (per unit WEBS)

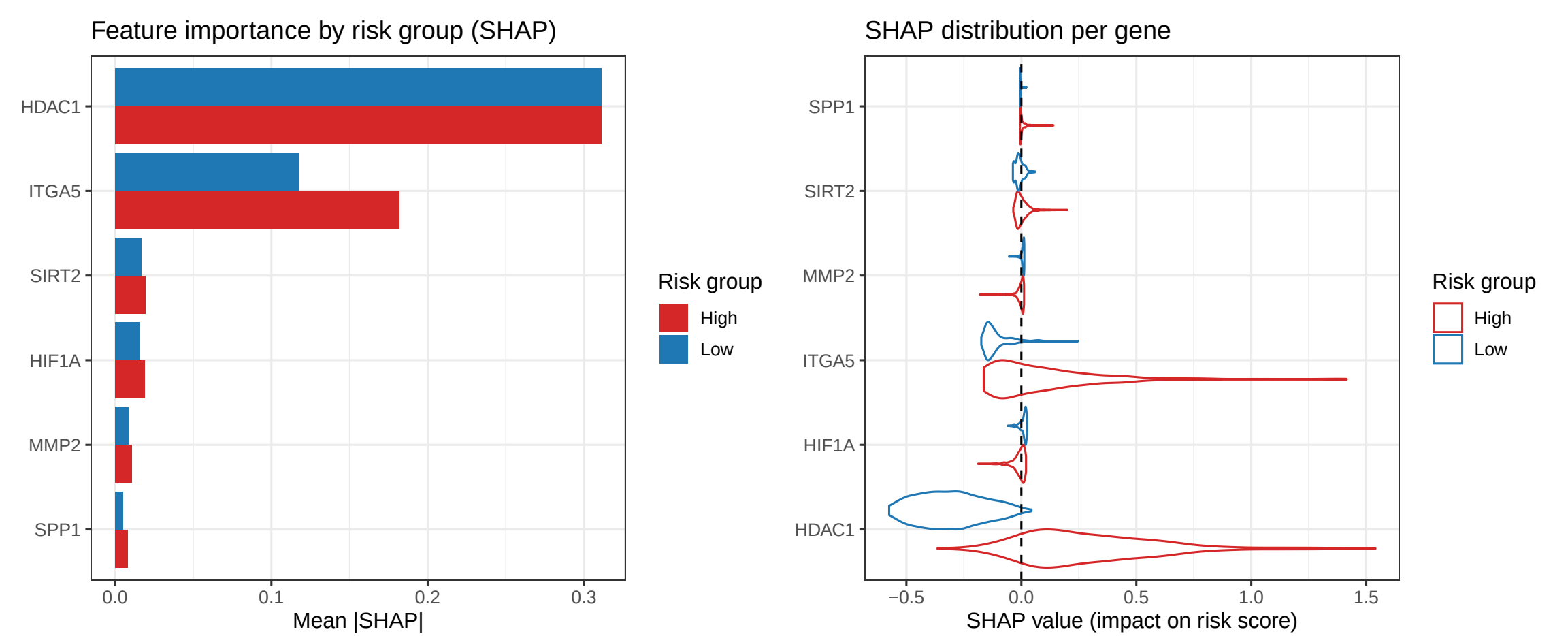

TCGA−GBM
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.65
0
30
60
90
120
Time (months)

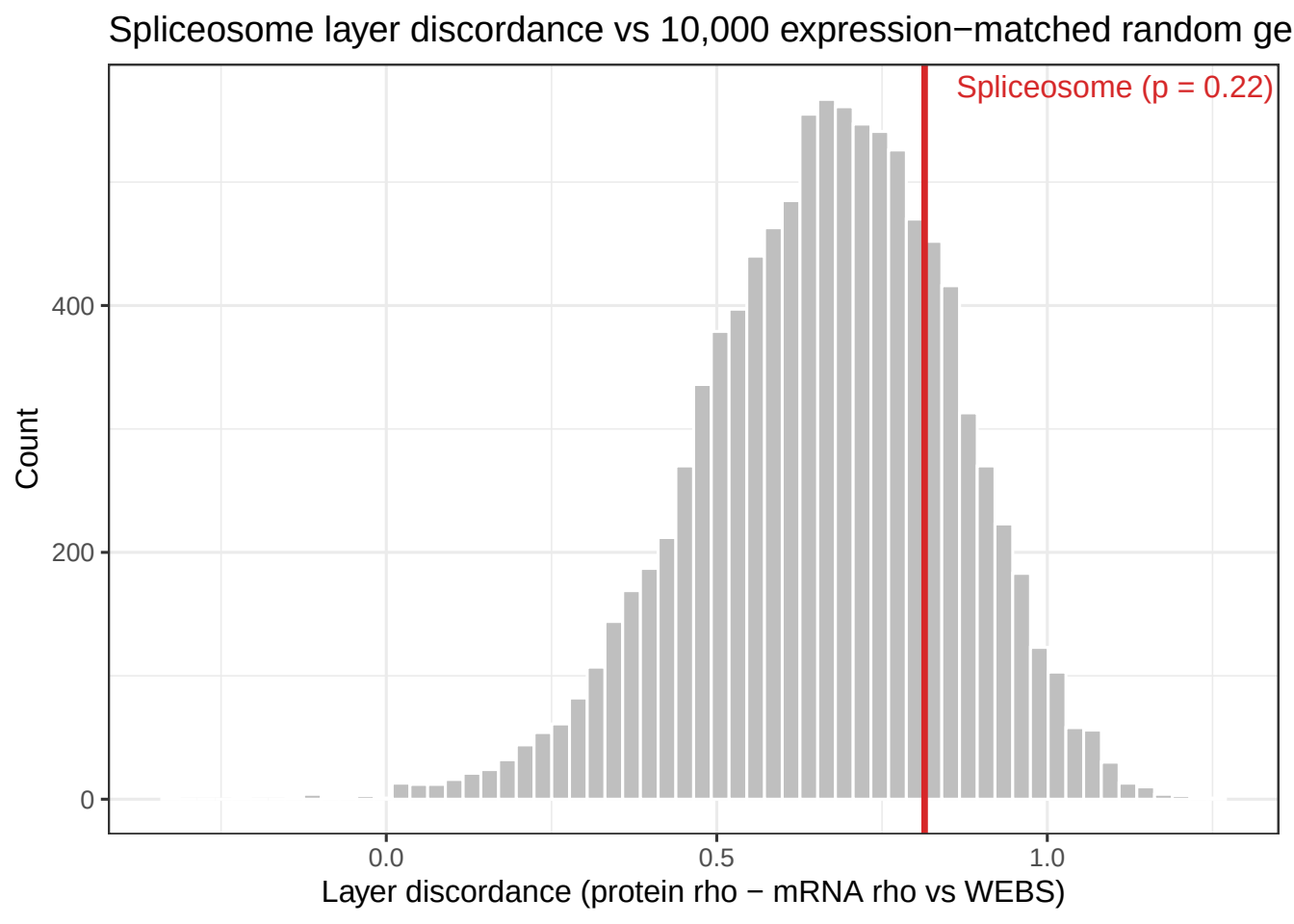

TCGA−LUAD
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.022
0
50
100
150
200
250
Time (months)

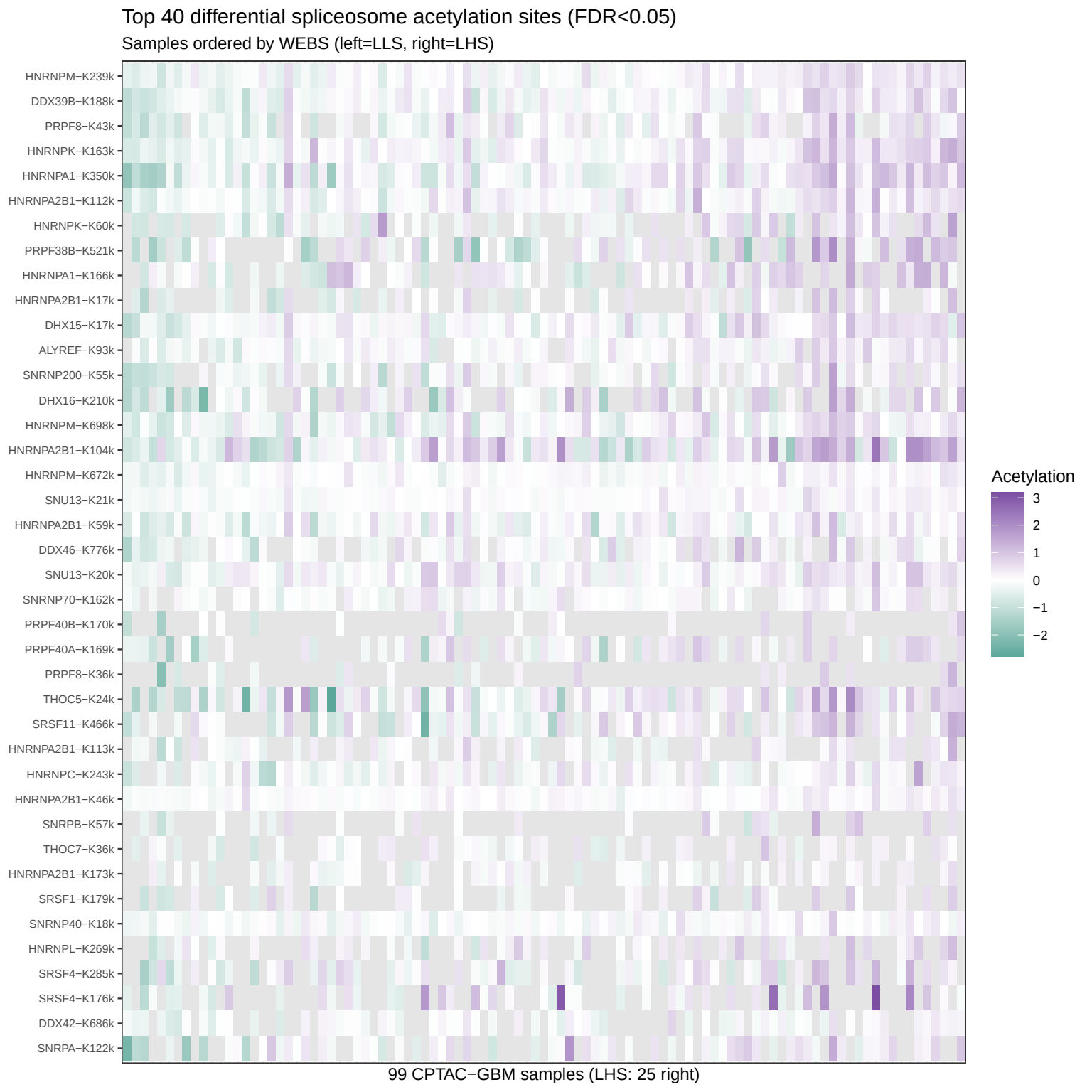

TCGA−BRCA
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.59
0
100
200
300
Time (months)

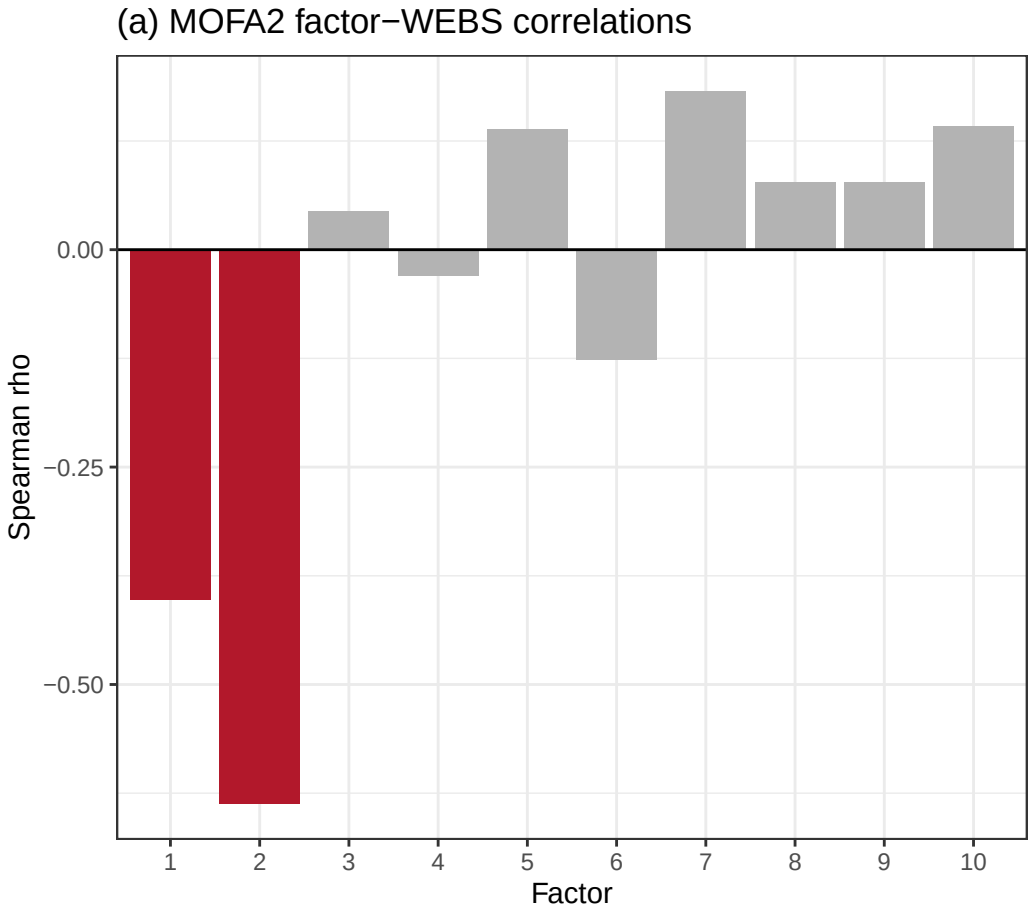

TCGA−SKCM
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.8
0
100
200
300
400
Time (months)

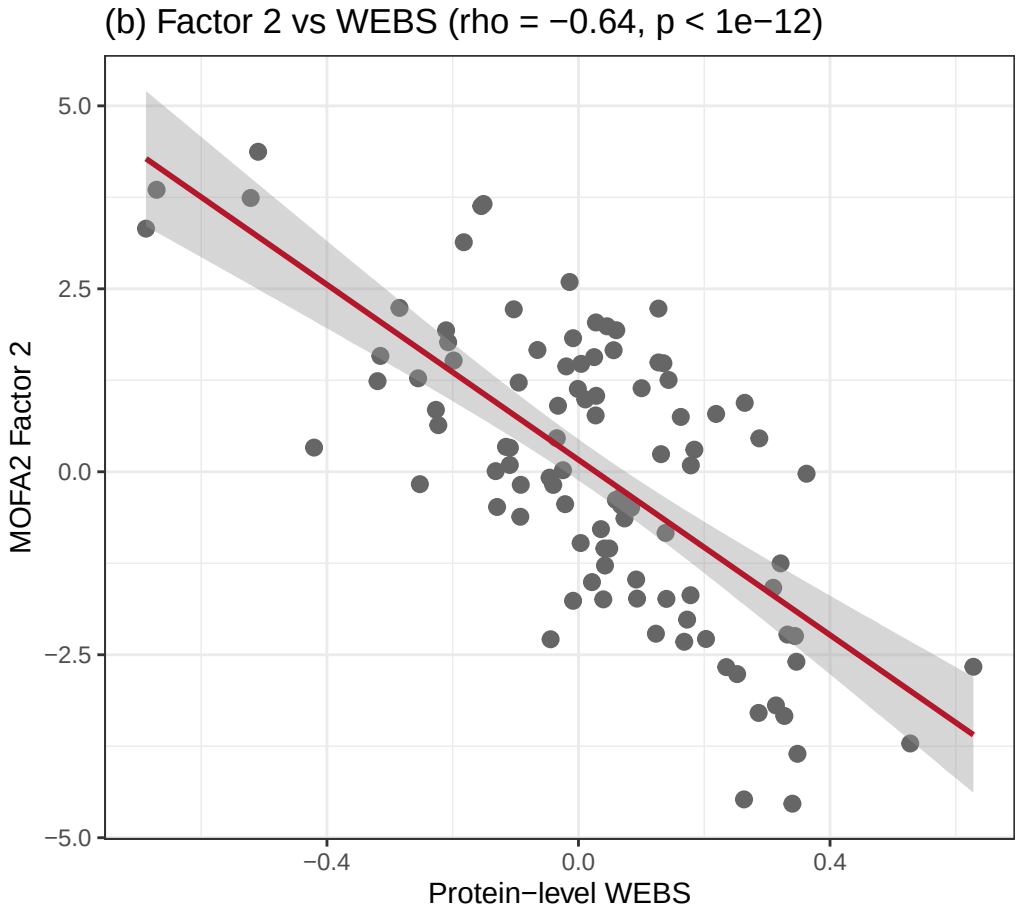

Bootstrap stability of LASSO−Cox feature selection
HDAC1
LDHB
VEGFA
KAT8
LDHA
SIRT1
KAT2A
KAT5
HDAC2
GAPDH
SIRT2
SPP1
ITGA5
SIRT6
ENO2
MMP2
EP300
SIRT7
CREBBP
FN1
SIRT3
CA9
HDAC3
PKM
KAT2B
HIF1A
ENO1
0.00
0.25
0.50
0.75
1.00
Selection frequency (1,000 bootstraps)
Not in final model
In final six−gene model

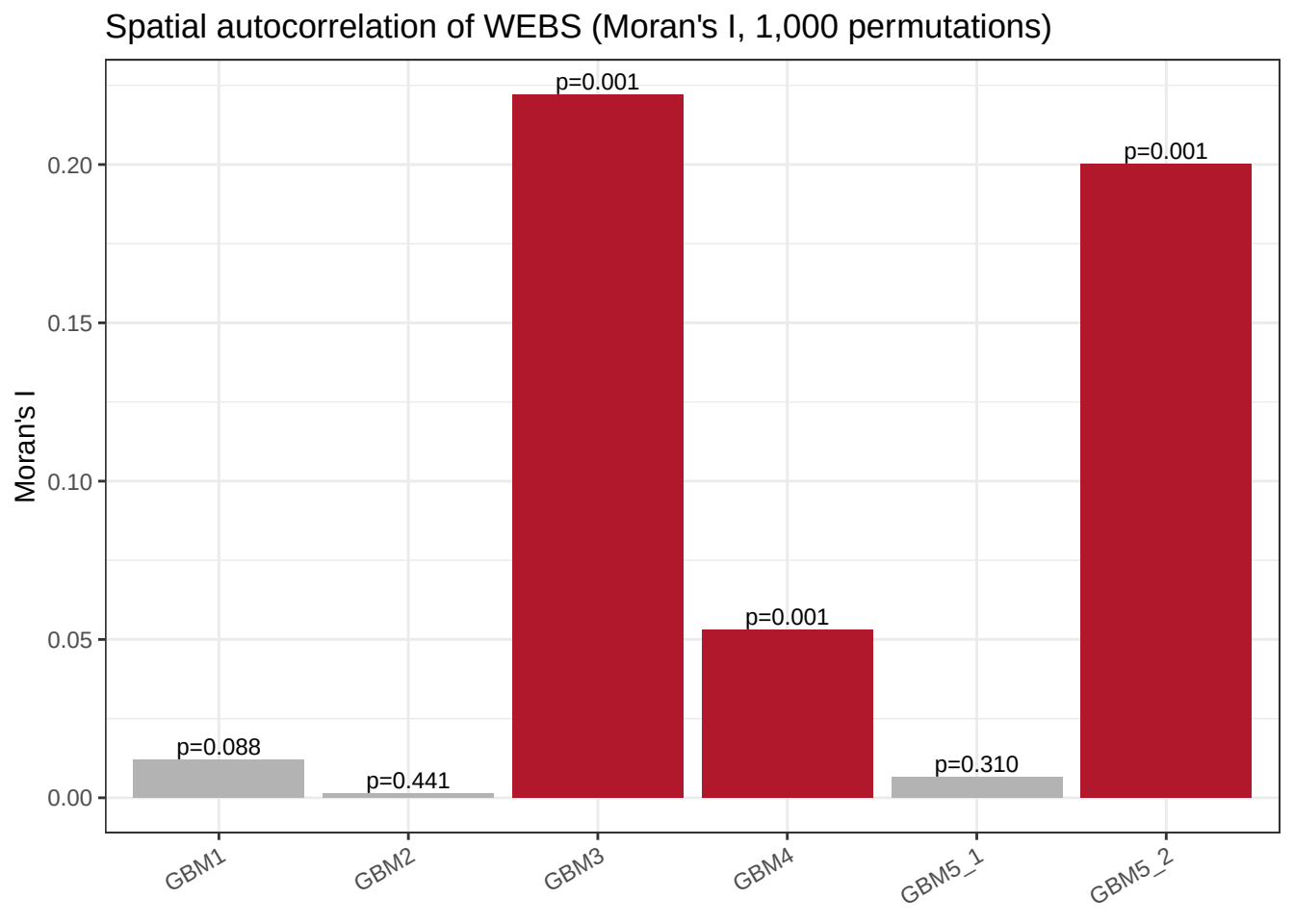

Feature importance by risk group (SHAP)
HDAC1
ITGA5
SIRT2
HIF1A
MMP2
SPP1
0.0
0.1
0.2
0.3
Mean |SHAP|
Risk group
High
Low
SHAP distribution per gene
SPP1
SIRT2
MMP2
ITGA5
HIF1A
HDAC1
−0.5
0.0
0.5
1.0
1.5
SHAP value (impact on risk score)
Risk group
High
Low

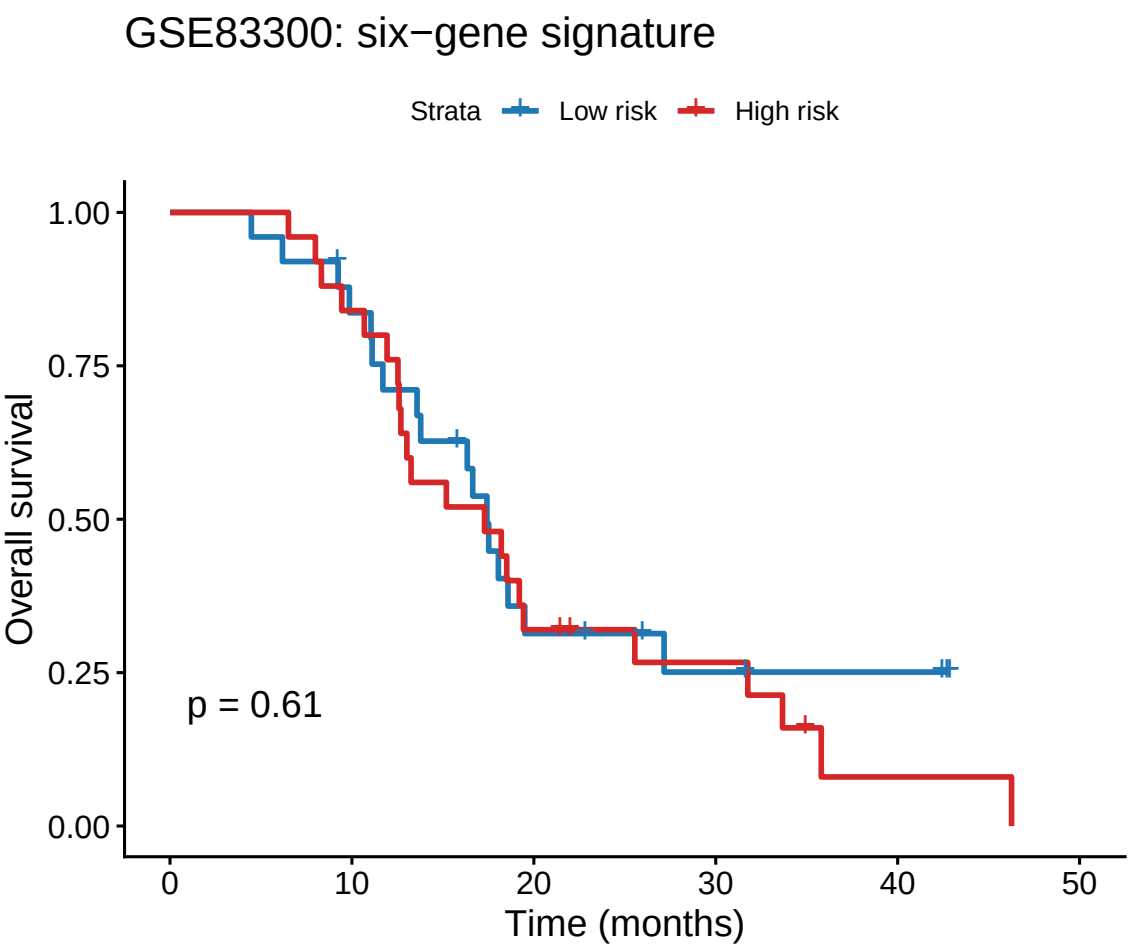

Spliceosome layer discordance vs 10,000 expression−matched random ge
Spliceosome (p = 0.22)
Count
400
200
0
0.0
0.5
1.0
Layer discordance (protein rho − mRNA rho vs WEBS)

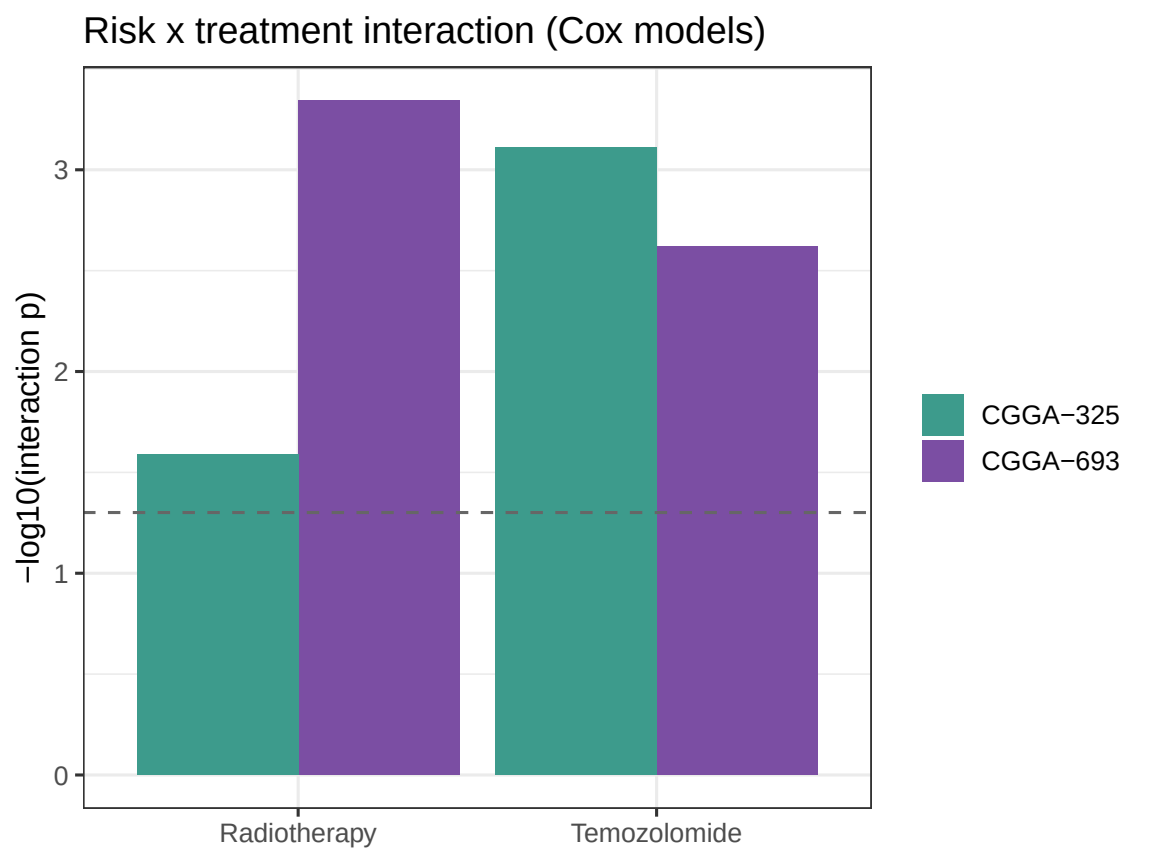

Top 40 differential spliceosome acetylation sites (FDR<0.05)
Samples ordered by WEBS (left=LLS, right=LHS)
HNRNPM−K239k
DDX39B−K188k
PRPF8−K43k
HNRNPK−K163k
HNRNPA1−K350k
HNRNPA2B1−K112k
HNRNPK−K60k
PRPF38B−K521k
HNRNPA1−K166k
HNRNPA2B1−K17k
DHX15−K17k
ALYREF−K93k
SNRNP200−K55k
DHX16−K210k
HNRNPM−K698k
HNRNPA2B1−K104k
HNRNPM−K672k
SNU13−K21k
HNRNPA2B1−K59k
DDX46−K776k
SNU13−K20k
SNRNP70−K162k
PRPF40B−K170k
PRPF40A−K169k
PRPF8−K36k
THOC5−K24k
SRSF11−K466k
HNRNPA2B1−K113k
HNRNPC−K243k
HNRNPA2B1−K46k
SNRPB−K57k
THOC7−K36k
HNRNPA2B1−K173k
SRSF1−K179k
SNRNP40−K18k
HNRNPL−K269k
SRSF4−K285k
SRSF4−K176k
DDX42−K686k
SNRPA−K122k
99 CPTAC−GBM samples (LHS: 25 right)
Acetylation
3
2
1
0
−1
−2

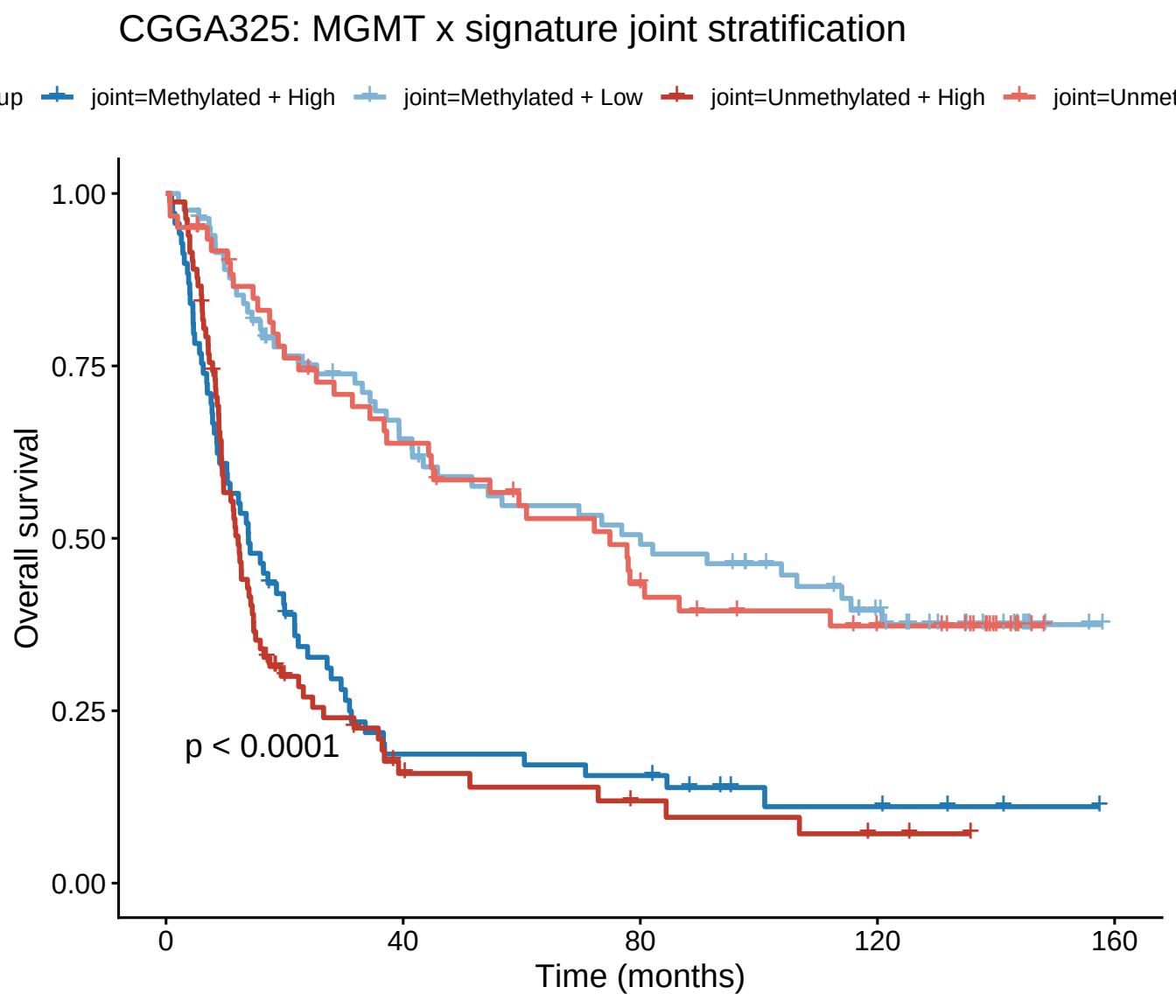

(a) MOFA2 factor−WEBS correlations
Spearman rho
0.00
−0.25
−0.50
1
2
3
4
5
6
7
8
9
10
Factor

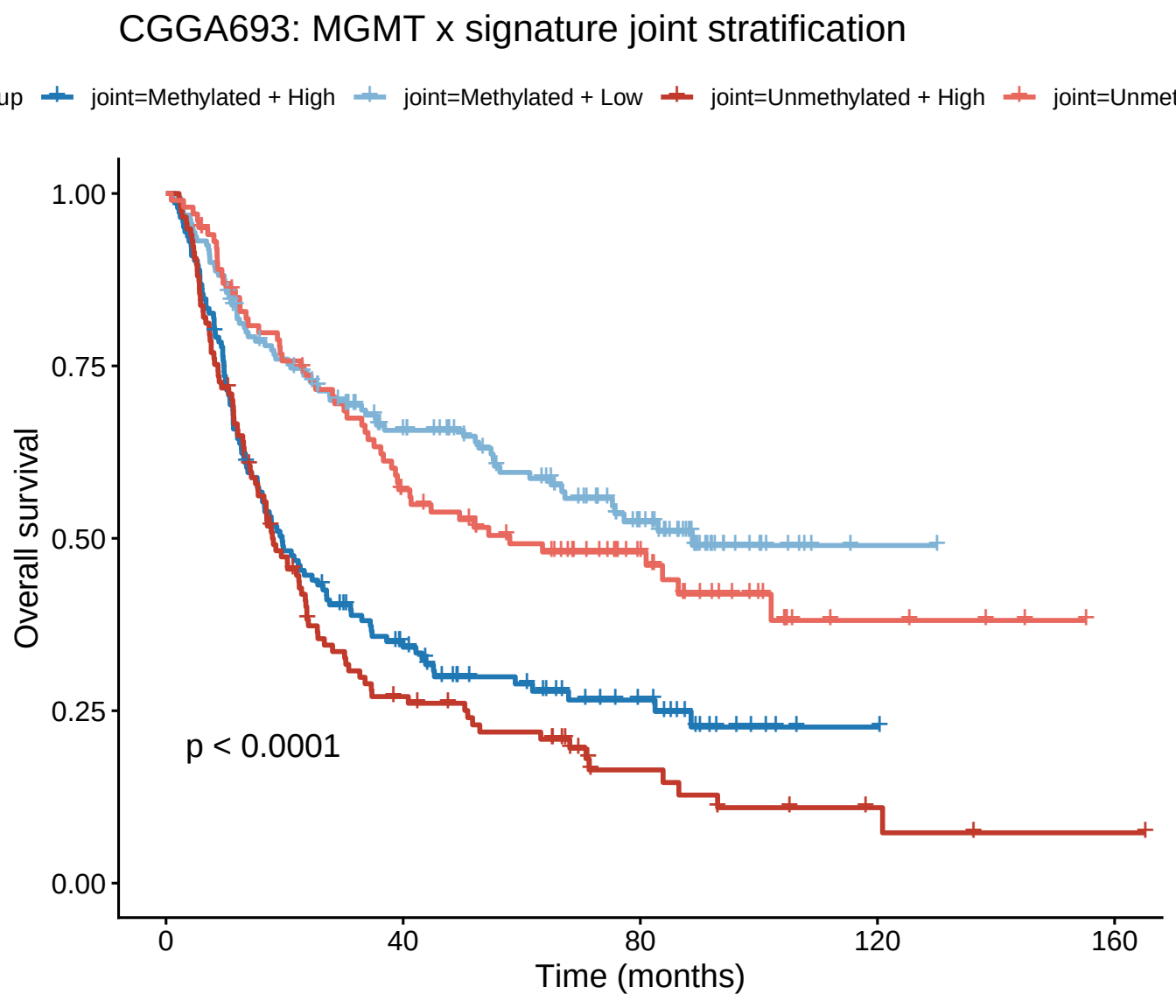

(b) Factor 2 vs WEBS (rho = −0.64, p < 1e−12)
MOFA2 Factor 2
5.0
2.5
0.0
−2.5
−5.0
−0.4
0.0
0.4
Protein−level WEBS

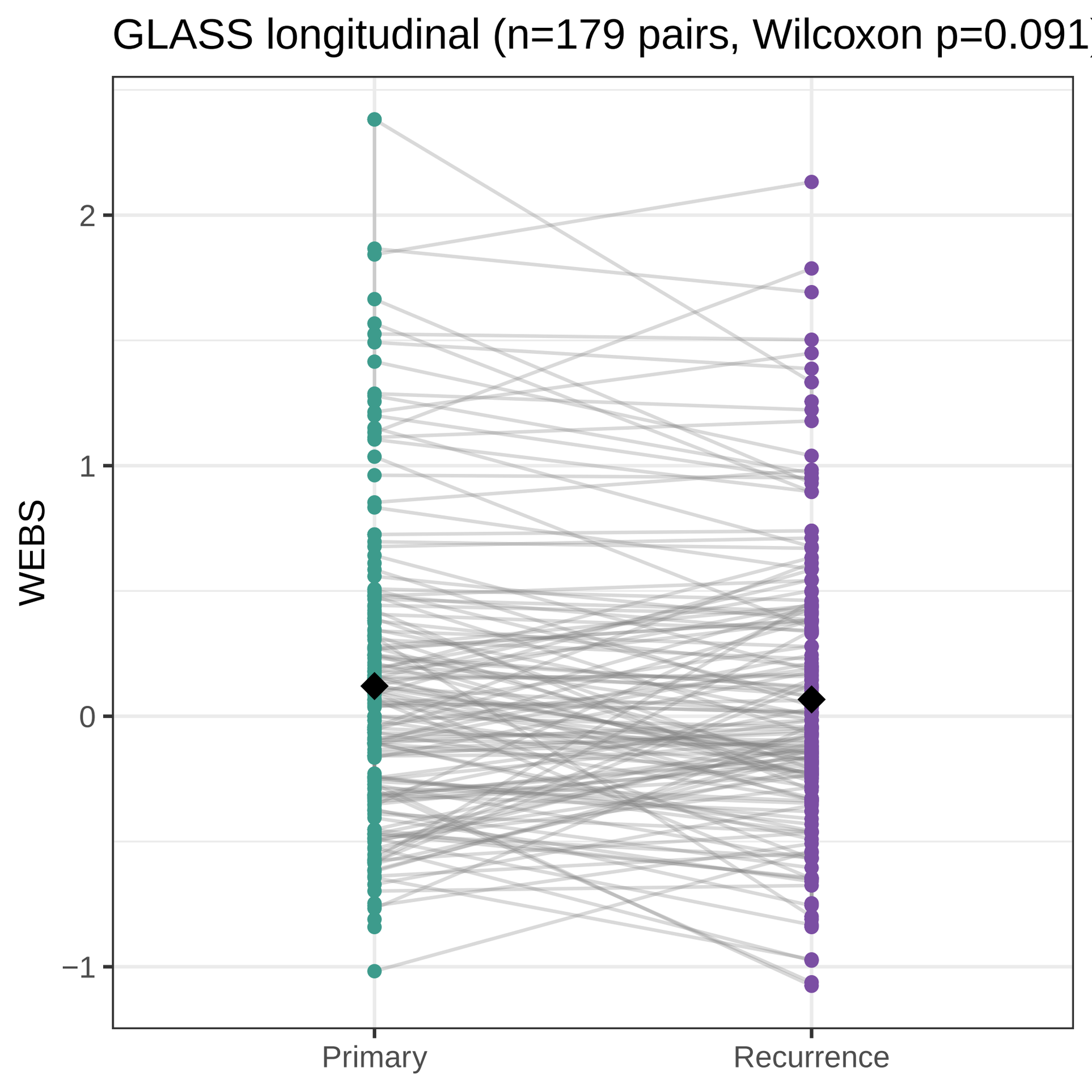

Spatial autocorrelation of WEBS (Moran's I, 1,000 permutations)
Moran's I
0.20
0.15
0.10
0.05
0.00
p=0.088
p=0.441
p=0.001
p=0.001
p=0.310
p=0.001
GBM1
GBM2
GBM3
GBM4
GBM5_1
GBM5_2

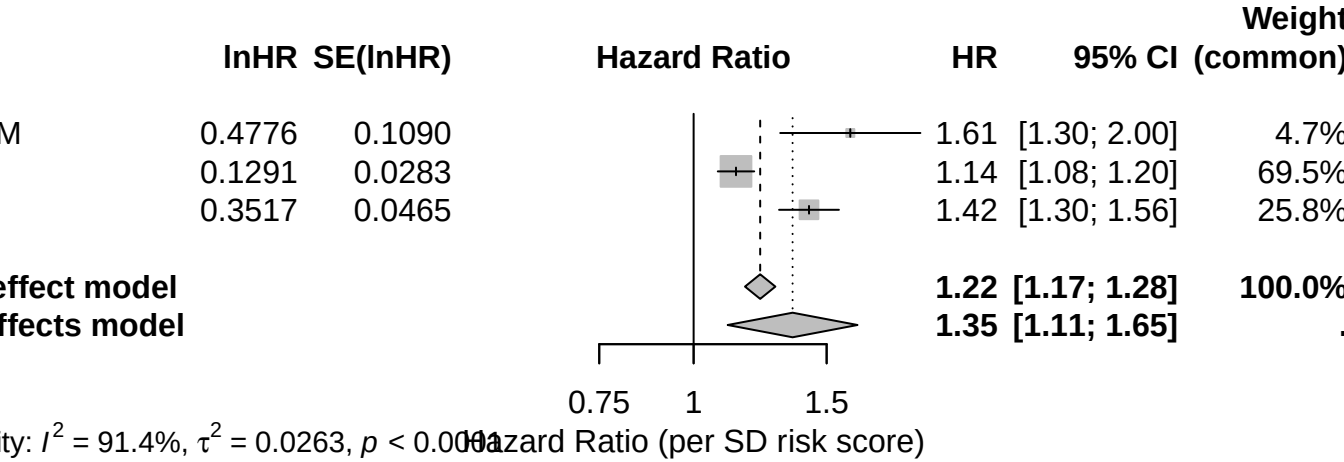

GSE83300: six−gene signature
Strata
Low risk
High risk
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.61
0
10
20
30
40
50
Time (months)

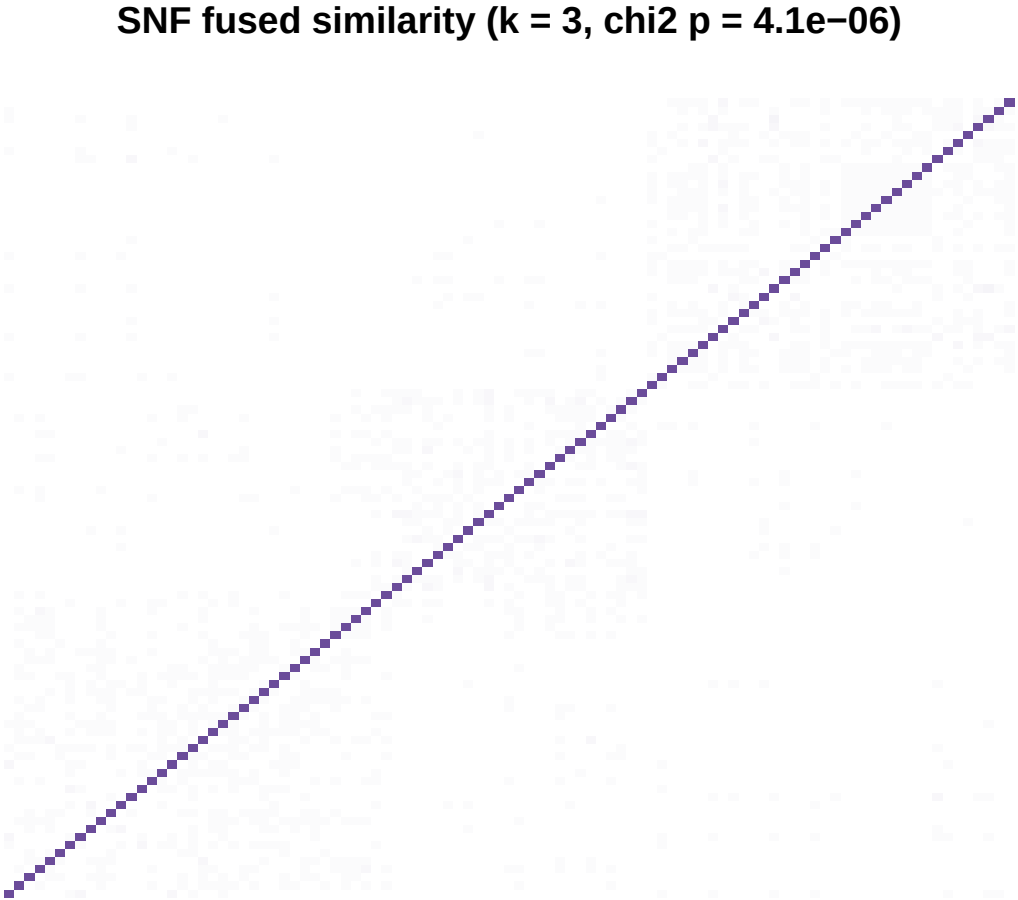

Risk x treatment interaction (Cox models)
−log10(interaction p)
0
1
2
3
Radiotherapy
Temozolomide
CGGA−325
CGGA−693

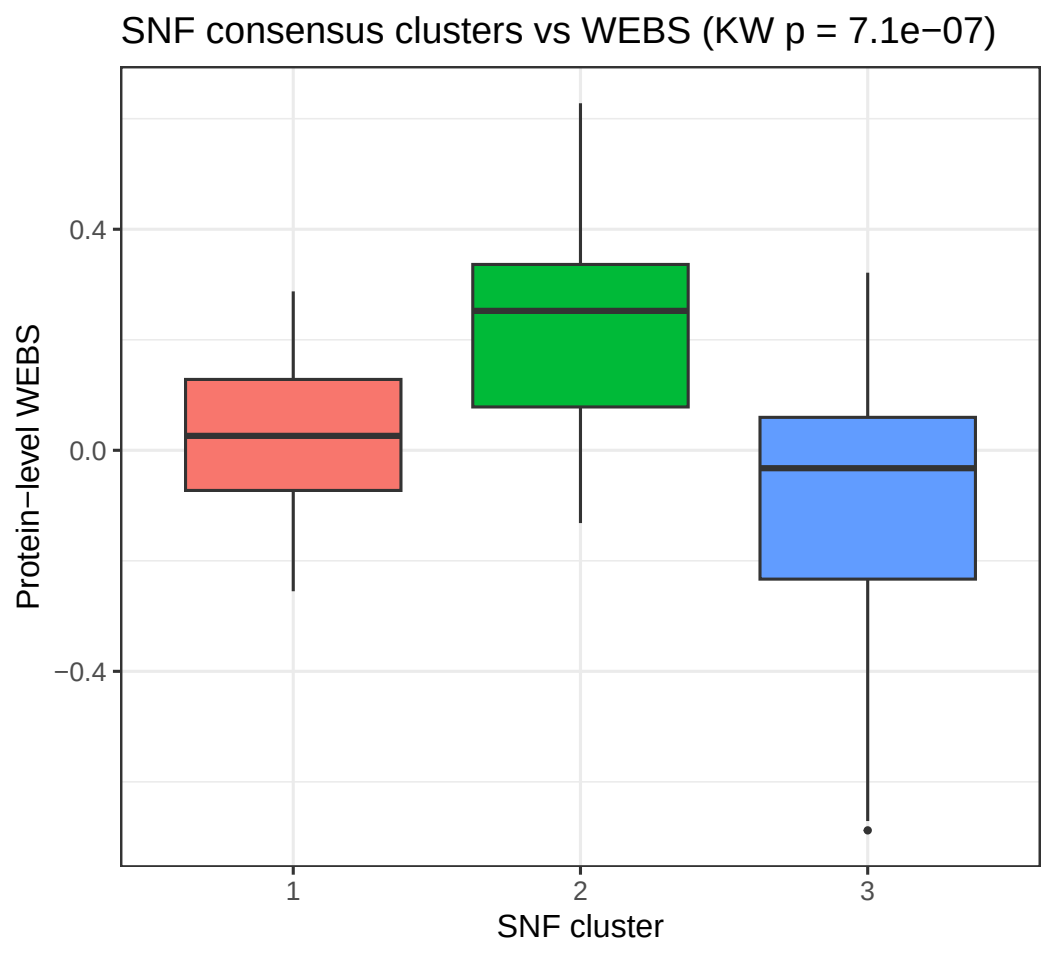

CGGA325: MGMT x signature joint stratification
up
joint=Methylated + High
joint=Methylated + Low
joint=Unmethylated + High
joint=Unmet
1.00
0.75
0.50
0.25
0.00
Overall survival
p < 0.0001
0
40
80
120
160
Time (months)

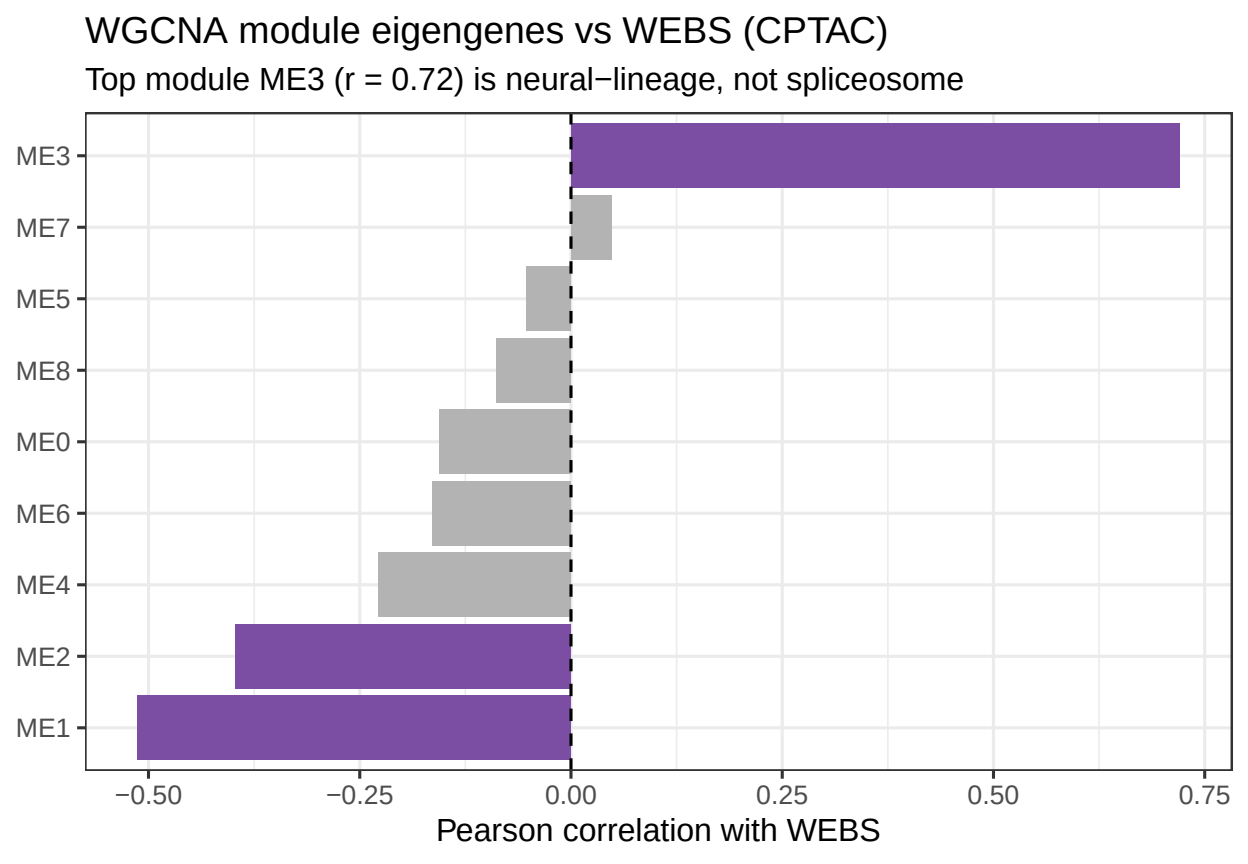

CGGA693: MGMT x signature joint stratification
up
joint=Methylated + High
joint=Methylated + Low
joint=Unmethylated + High
joint=Unmet
1.00
0.75
0.50
0.25
0.00
Overall survival
p < 0.0001
0
40
80
120
160
Time (months)


GLASS longitudinal (n=179 pairs, Wilcoxon p=0.091)
WEBS
2
1
0
−1
Primary
Recurrence

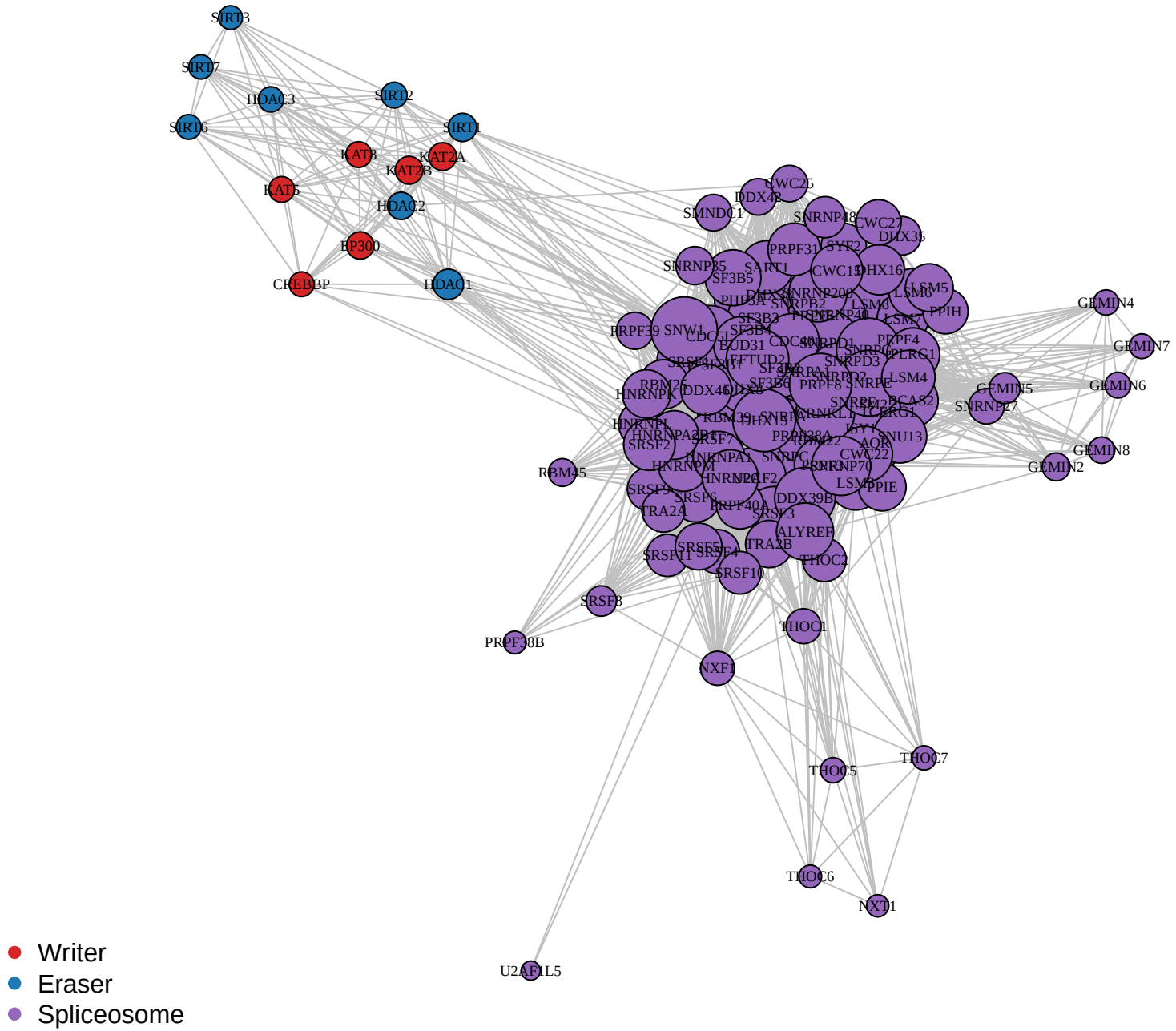

lnHR
SE(lnHR)
Hazard Ratio
HR
95% CI
Weight (common)
M
0.4776
0.1090
1.61
[1.30; 2.00]
4.7%
0.1291
0.0283
1.14
[1.08; 1.20]
69.5%
0.3517
0.0465
1.42
[1.30; 1.56]
25.8%
effect model
1.22
[1.17; 1.28]
100.0%
ffects model
1.35
[1.11; 1.65]
0.75
1
1.5
ity: I^2 = 91.4%, τ^2 = 0.0263, p < 0.0001
Hazard Ratio (per SD risk score)

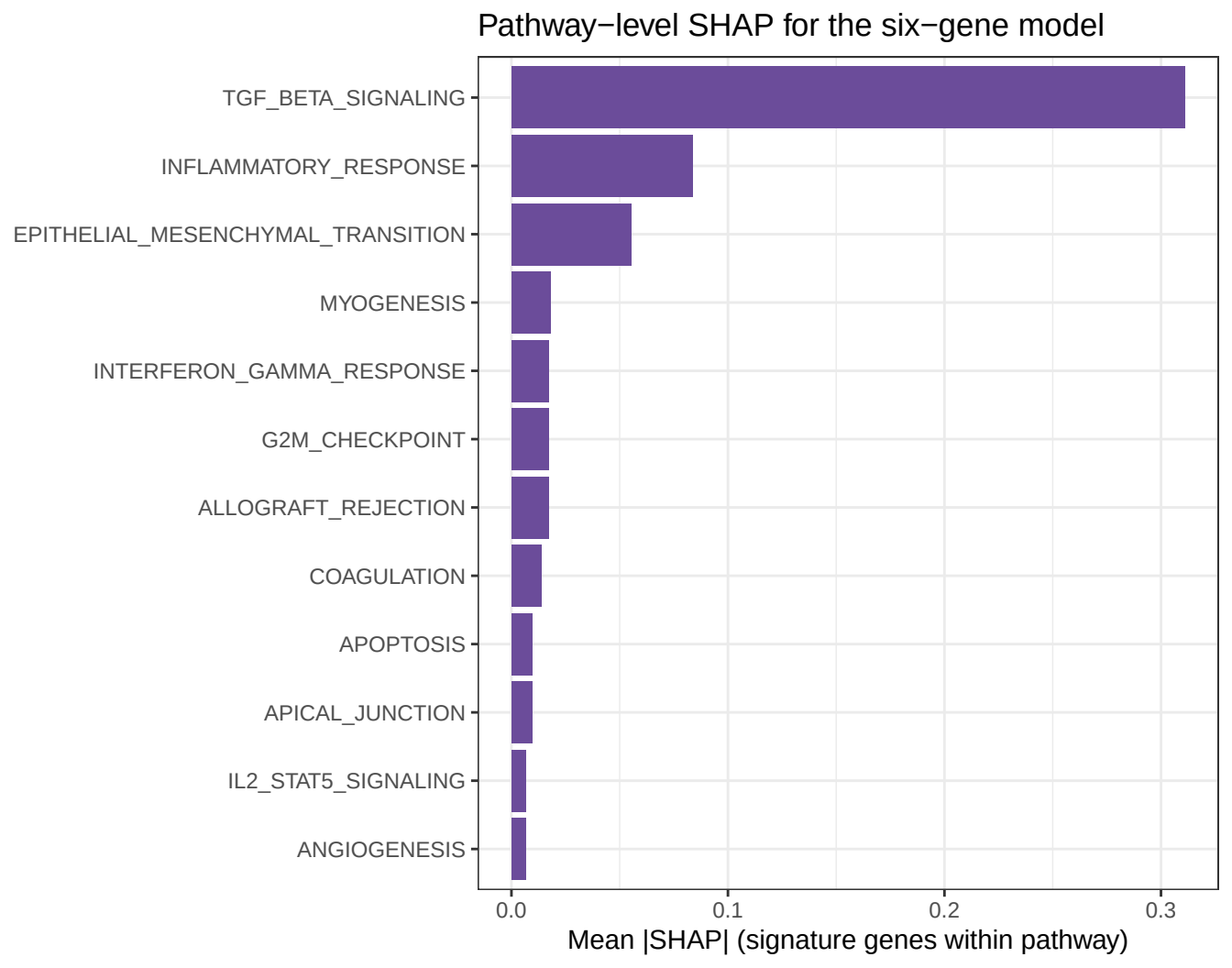

SNF fused similarity (k = 3, chi2 p = 4.1e−06)

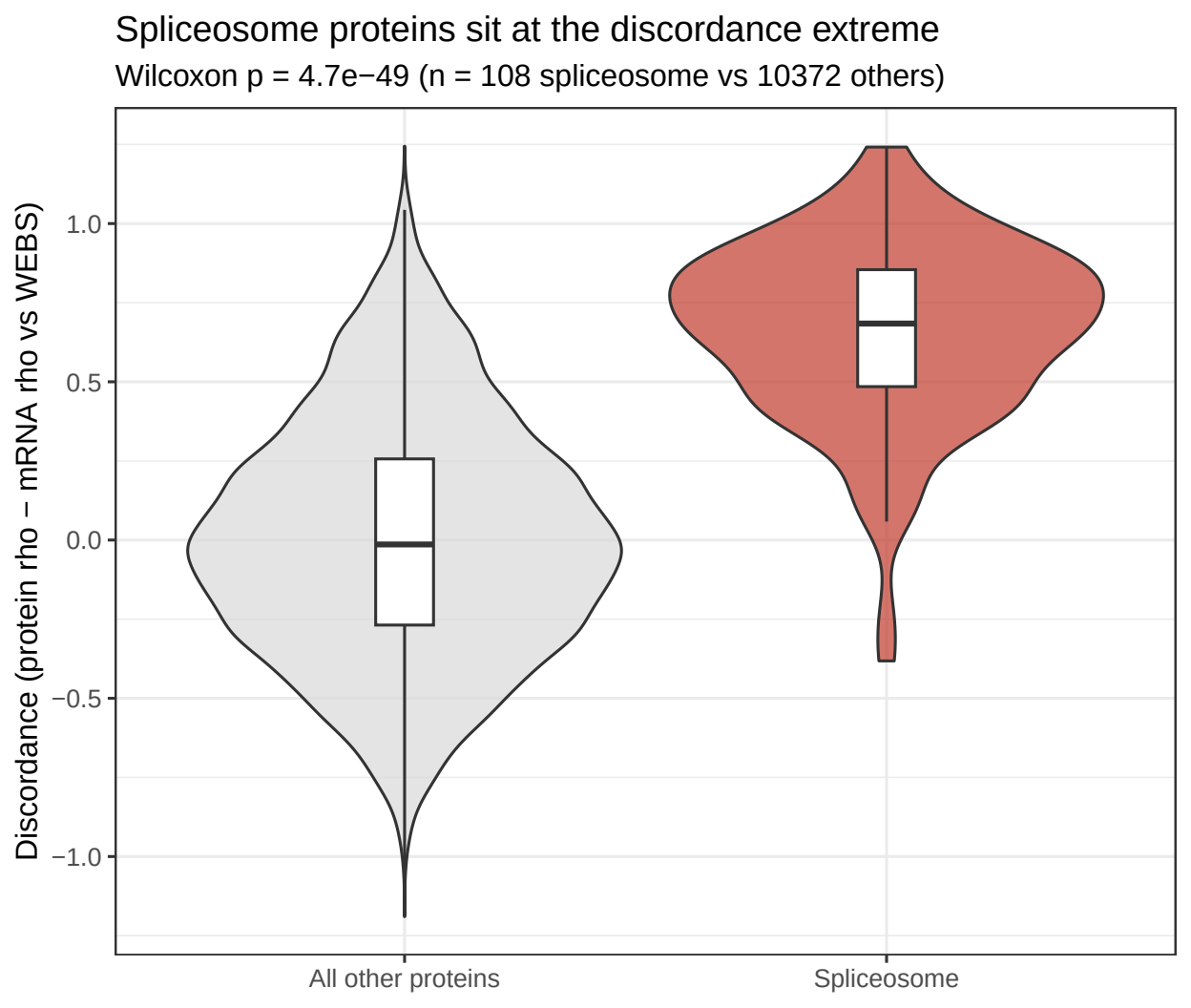

SNF consensus clusters vs WEBS (KW p = 7.1e−07)
Protein−level WEBS
0.4
0.0
−0.4
1
2
3
SNF cluster

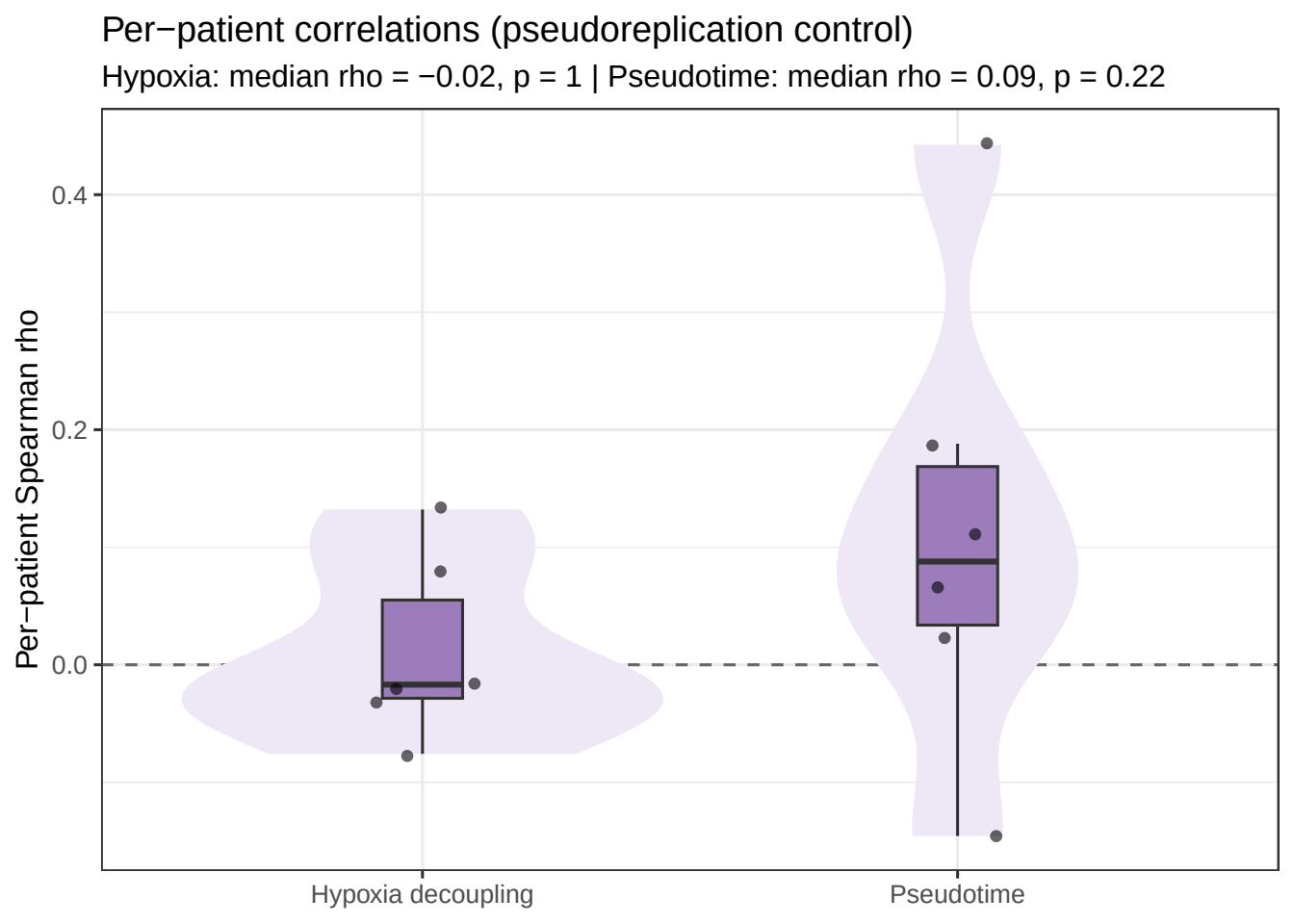

WGCNA module eigengenes vs WEBS (CPTAC)
Top module ME3 (r = 0.72) is neural−lineage, not spliceosome
ME3
ME7
ME5
ME8
ME0
ME6
ME4
ME2
ME1
−0.50
−0.25
0.00
0.25
0.50
0.75
Pearson correlation with WEBS

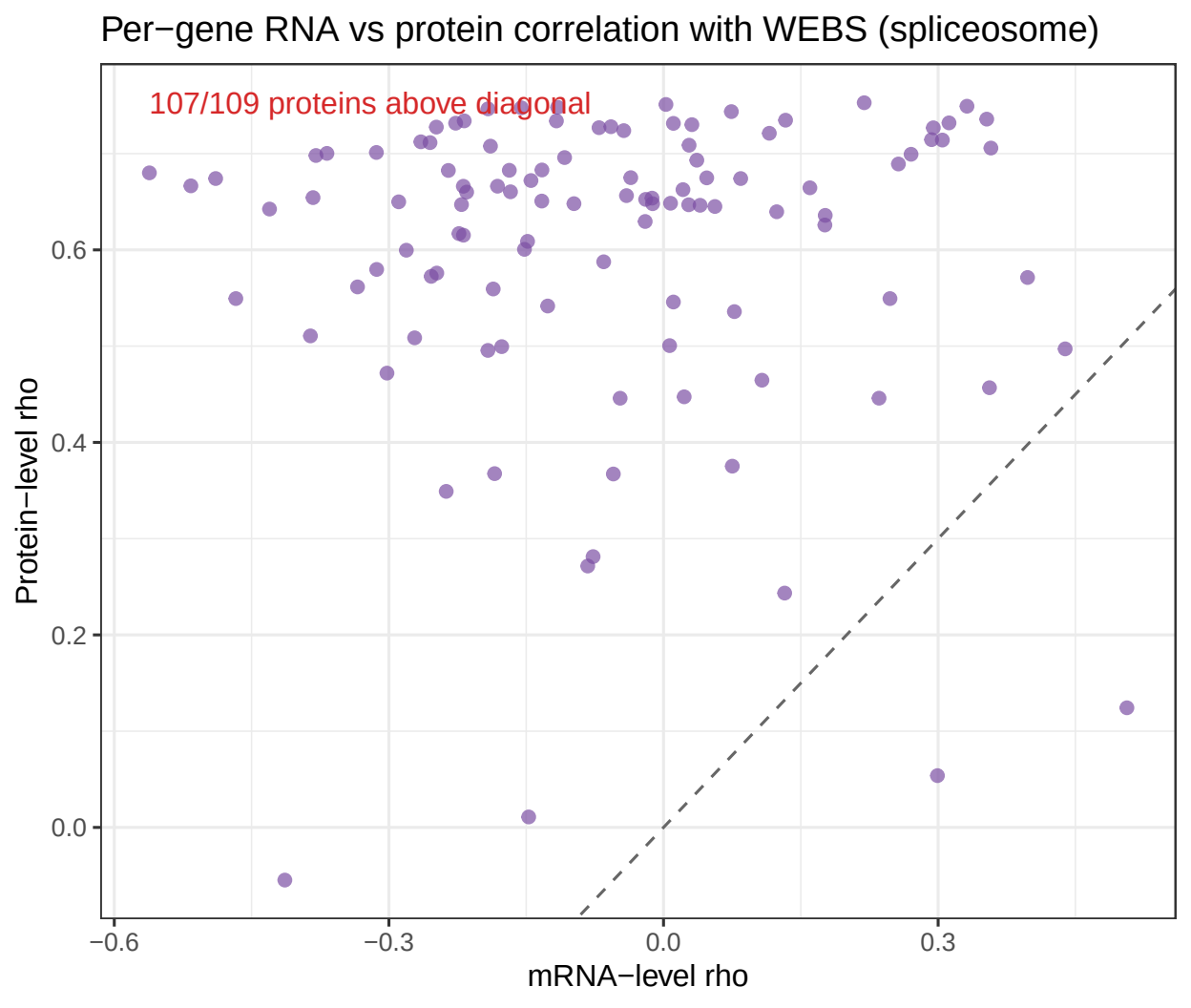
Writer−Eraser−Spliceosome PPI (121 nodes, 3349 edges)

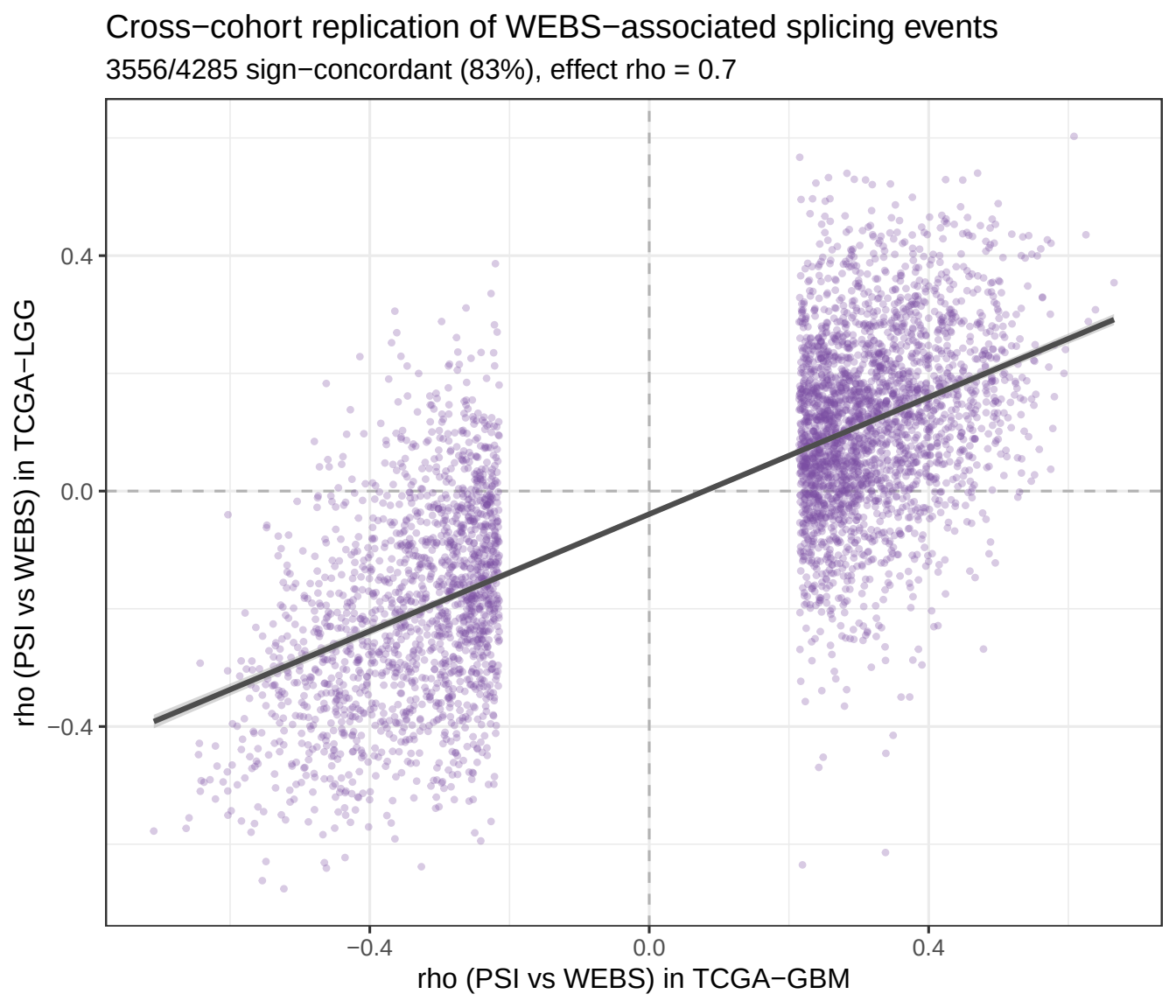



Pathway−level SHAP for the six−gene model
TGF_BETA_SIGNALING
INFLAMMATORY_RESPONSE
EPITHELIAL_MESENCHYMAL_TRANSITION
MYOGENESIS
INTERFERON_GAMMA_RESPONSE
G2M_CHECKPOINT
ALLOGRAFT_REJECTION
COAGULATION
APOPTOSIS
APICAL_JUNCTION
IL2_STAT5_SIGNALING
ANGIOGENESIS
0.0
0.1
0.2
0.3
Mean |SHAP| (signature genes within pathway)


Spliceosome proteins sit at the discordance extreme
Wilcoxon p = 4.7e−49 (n = 108 spliceosome vs 10372 others)
Discordance (protein rho − mRNA rho vs WEBS)
1.0
0.5
0.0
−0.5
−1.0
All other proteins
Spliceosome


Per−patient correlations (pseudoreplication control)
Hypoxia: median rho = −0.02, p = 1 | Pseudotime: median rho = 0.09, p = 0.22
Per−patient Spearman rho
0.4
0.2
0.0
Hypoxia decoupling
Pseudotime


Per−gene RNA vs protein correlation with WEBS (spliceosome)
107/109 proteins above diagonal
Protein−level rho
mRNA−level rho
0.6
0.4
0.2
0.0
−0.6
−0.3
0.0
0.3


Cross−cohort replication of WEBS−associated splicing events
3556/4285 sign−concordant (83%), effect rho = 0.7
rho (PSI vs WEBS) in TCGA−LGG
rho (PSI vs WEBS) in TCGA−GBM
0.4
0.0
−0.4
−0.4
0.0
0.4